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Air conditioning system, abnormality estimation method for air conditioning system, air conditioner, and abnormality estimation method for air conditioner

Abstract

An air conditioner includes a refrigerant circuit in which indoor units are connected to an outdoor unit. The conditioner includes a processor to detect a state quantity related to control on the air conditioner, acquire a detected value of the detected quantity and estimate occurrence of abnormality of the circuit by using a detected value of a feature value by assuming that the quantity related to abnormality of the circuit is adopted as the feature value. The estimating adopts the outdoor unit and each of the indoor units as a single pair, estimates occurrence of abnormality in the circuit for each of the pairs, estimates that abnormality has occurred in the indoor unit of a subject pair when estimating that abnormality has occurred in any of the pair, and estimates that abnormality has occurred in the outdoor unit when estimating that abnormality has occurred in all of the pairs.

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Background/Summary

FIELD

(1) The present invention relates to an air conditioning system, an abnormality estimation method for an air conditioning system, an air conditioner, and an abnormality estimation method for an air conditioner.

BACKGROUND

(2) Various methods for detecting abnormality related to a refrigerant circuit in an air conditioner or a sign of the abnormality have been proposed. For example, Patent Literature 1 proposes a method of detecting abnormality of an air conditioner by performing learning while adopting, as normal data, an ability score that is obtained by using values detected by various kinds of sensors, and comparing values that are detected by the various kinds of sensors in a period different from a learning period with the normal data.

CITATION LIST

Patent Literature

(3) Patent Literature 1: Japanese Laid-open Patent Publication No. 2020-16358

SUMMARY

Technical Problem

(4) However, in Patent Literature 1, only occurrence of abnormality of the air conditioner is estimated. Therefore, for example, with respect to an air conditioner in which a plurality of indoor units are connected to an outdoor unit by refrigerant pipes, it is impossible to estimate in which of the outdoor unit and the indoor units abnormality has occurred.

(5) In view of the foregoing situations, an object of the present invention is to provide an air conditioning system, an abnormality estimation method for an air conditioning system, an air conditioner, and an abnormality estimation method for an air conditioner, which are able to estimate in which of an indoor unit and an outdoor unit abnormality has occurred.

Solution to Problem

(6) According to an aspect of an embodiment, an air conditioning system includes an air conditioner and a server. The air conditioner includes a refrigerant circuit in which at least one or more indoor units are connected to an outdoor unit by refrigerant pipes. The server is communicably connected to the air conditioner. The air conditioner includes a first memory and first processing circuitry coupled to the first memory. The first processing circuitry is configured to detect a state quantity related to control on the air conditioner. The first processing circuitry is configured to acquire a detected value of the state quantity that is detected by the detecting. The first processing circuitry is configured to transmit the detected value acquired by the acquiring to the server. The server includes a second memory and second processing circuitry coupled to the second memory. The second processing circuitry is configured to receive the detected value from the air conditioner. The second processing circuitry is configured to estimate occurrence of abnormality of the refrigerant circuit by using a detected value of a feature value by assuming that the state quantity related to abnormality of the refrigerant circuit is adopted as the feature value. The estimating adopts the outdoor unit and each of the indoor units as a single pair, estimates occurrence of abnormality in the refrigerant circuit for each of the pairs, estimates that abnormality has occurred in the indoor unit of a subject pair when estimating that abnormality has occurred in any of the pairs, and estimates that abnormality has occurred in the outdoor unit when estimating that abnormality has occurred in all of the pairs

Advantageous Effects of Invention

(7) According to one aspect, it is possible to estimate in which of an outdoor unit and an indoor unit abnormality has occurred.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is an explanatory diagram illustrating an example of an air conditioner of a present embodiment.
- (2) FIG. 2 is an explanatory diagram illustrating an example of an outdoor unit and indoor units.
- (3) FIG. 3 is a block diagram illustrating one example of a control circuit of the outdoor unit.
- (4) FIG. 4 is an explanatory diagram illustrating an example of pairing such that the outdoor unit and each of the indoor units are combined as a single pair.
- (5) FIG. 5 is a Mollier diagram illustrating a state of a change in a refrigerant of the air conditioner.
- (6) FIG. 6 is an explanatory diagram illustrating examples of a first feature value that is used for first to third cooler estimation models and a second feature value that is used for a cooling-period abnormality estimation model.
- (7) FIG. 7 is an explanatory diagram illustrating examples of a first feature value that is used for first to third heater estimation models and a second feature value that is used for a heating-period abnormality estimation model.
- (8) FIG. 8A is an explanatory diagram illustrating an example in which an estimation result obtained by the first cooler estimation model and an estimation result obtained by the second cooler estimation model are not interpolated by a sigmoid curve.
- (9) FIG. 8B is an explanatory diagram illustrating an example in which the estimation result obtained by the first cooler estimation model and the estimation result obtained by the second cooler estimation model are interpolated by a sigmoid curve.
- (10) FIG. 9A is an explanatory diagram illustrating an example in which an estimation result obtained by the first heater estimation model and an estimation result obtained by the second heater estimation model are not interpolated by a sigmoid curve.
- (11) FIG. 9B is an explanatory diagram illustrating an example in which the estimation result obtained by the first heater estimation model and the estimation result obtained by the second heater estimation model are interpolated by a sigmoid curve.
- (12) FIG. 10 is an explanatory diagram illustrating an example of the way of distribution of detected values of the second feature values of the abnormality estimation model.
- (13) FIG. 11 is an explanatory diagram illustrating an example of abnormality detection by an outlier.
- (14) FIG. 12 is an explanatory diagram illustrating an example of a determination result obtained by a determination unit.
- (15) FIG. 13 is a flowchart illustrating an example of processing operation performed by a control circuit related to an estimation process.
- (16) FIG. 14 is a flowchart illustrating an example of processing operation performed by the control circuit related to a remaining refrigerant amount estimation process.
- (17) FIG. 15 is an explanatory diagram illustrating an example of an air conditioning system of a second embodiment.

DESCRIPTION OF EMBODIMENTS

(18) Embodiments of an air conditioning system, an abnormality estimation method for an air conditioning system, an air conditioner, and an abnormality estimation method for an air conditioner disclosed in the present application will be described in detail below based on the drawings. Meanwhile, the disclosed technology is not limited by the present embodiments. In addition, each of the embodiments described below may be modified appropriately as long as no

contradiction is derived.

First Embodiment

(19) Configuration of Air Conditioner

(20) FIG. 1 is an explanatory diagram illustrating an example of an air conditioner **1** of the present embodiment. The air conditioner **1** illustrated in FIG. 1 includes a single outdoor unit **2** and N indoor units **3** (N is a natural number equal to or larger than 2). The outdoor unit **2** is connected to each of the indoor units **3** in a parallel manner via a liquid pipe **4** and a gas pipe **5**. Further, a refrigerant circuit **6** of the air conditioner **1** is formed by connecting the outdoor unit **2** and the indoor units **3** to each other by refrigerant pipes, such as the liquid pipe **4** and the gas pipe **5**.

(21) Configuration of Outdoor Unit

(22) FIG. 2 is an explanatory diagram illustrating an example of the outdoor unit **2** and the N indoor units **3**. The outdoor unit **2** includes a compressor **11**, a four-way valve **12**, an outdoor heat exchanger **13**, an outdoor unit expansion valve **14**, a first stop valve **15**, a second stop valve **16**, an accumulator **17**, an outdoor unit fan **18**, and a control circuit **19**. The compressor **11**, the four-way valve **12**, the outdoor heat exchanger **13**, the outdoor unit expansion valve **14**, the first stop valve **15**, the second stop valve **16**, and the accumulator **17** as described above are connected to one another by each of refrigerant pipes to be described in detail below, and used to form an outdoor-side refrigerant circuit that is a part of the refrigerant circuit **6**.

(23) The compressor **11** is, for example, a variable capacity compressor of a pressurized container type in which operating capacity can be changed in accordance with drive of a motor (not illustrated) for which rotation speed is controlled by an inverter. A refrigerant discharge side of the compressor **11** is connected to a first port **12A** of the four-way valve **12** by a discharge pipe **21**. Further, a refrigerant suction side of the compressor **11** is connected to a refrigerant outflow side of the accumulator **17** by a suction pipe **22**.

(24) The four-way valve **12** is a valve for changing a flow direction of a refrigerant in the refrigerant circuit **6**, and includes the first to fourth ports **12A** to **12D**. The first port **12A** is connected to the refrigerant discharge side of the compressor **11** by the discharge pipe **21**. The second port **12B** is connected to one refrigerant port of the outdoor heat exchanger **13** by an outdoor refrigerant pipe **23**. The third port **12C** is connected to a refrigerant inflow side of the accumulator **17** by an outdoor refrigerant pipe **26**. Further, the fourth port **12D** is connected to the second stop valve **16** by an outdoor gas pipe **24**.

(25) The outdoor heat exchanger **13** performs heat exchange between the refrigerant and outdoor air that is taken into the outdoor unit **2** by rotation of the outdoor unit fan **18**. The one refrigerant port of the outdoor heat exchanger **13** is connected to the second port **12B** of the four-way valve **12** by the outdoor refrigerant pipe **23**. Another refrigerant ports of the outdoor heat exchanger **13** is connected to the first stop valve **15** by an outdoor liquid pipe **25**. The outdoor heat exchanger **13** functions as a condenser when the air conditioner **1** performs cooling operation, and functions as an evaporator when the air conditioner **1** performs heating operation.

(26) The outdoor unit expansion valve **14** is an electronic expansion valve that is arranged in the outdoor liquid pipe **25** and that is driven by a pulse motor (not illustrated). A degree of opening of the outdoor unit expansion valve **14** is adjusted in accordance with the number of pulses given to the pulse motor, so that an amount of the refrigerant that flows into the outdoor heat exchanger **13** or an amount of the refrigerant that flows out of the outdoor heat exchanger **13** is adjusted. The degree of opening of the outdoor unit expansion valve **14** is adjusted such that when the air conditioner **1** performs heating operation, a degree of suction superheat of refrigerant at the refrigerant suction side of the compressor **11** reaches a target suction superheat. Further, the degree of opening of the outdoor unit expansion valve **14** is set to a fully-opened state when the air conditioner **1** performs cooling operation.

(27) The refrigerant inflow side of the accumulator **17** is connected to the third port **12C** of the four-way valve **12** by the outdoor refrigerant pipe **26**. Further, the refrigerant outflow side of the

accumulator **17** is connected to a refrigerant inflow side of the compressor **11** by the suction pipe **22**. The accumulator **17** separates the refrigerant that has flown into the accumulator **17** from the outdoor refrigerant pipe **26** into gas refrigerant and liquid refrigerant, and causes only the gas refrigerant to be sucked into the compressor **11**.

(28) The outdoor unit fan **18** is made of a resin material and arranged in the vicinity of the outdoor heat exchanger **13**. The outdoor unit fan **18** takes outdoor air into the outdoor unit **2** from a suction port (not illustrated) in accordance with rotation of a fan motor (not illustrated), and discharges the outdoor air, which has been subjected to heat exchange with the refrigerant in the outdoor heat exchanger **13**, to the outside of the outdoor unit **2** from a discharge port (not illustrated).

(29) Further, a plurality of sensors are arranged in the outdoor unit **2**. In the discharge pipe **21**, a discharge pressure sensor **31** that detects discharge pressure as pressure of the refrigerant that is discharged from the compressor **11**, and a discharge temperature sensor **32** that detects temperature of the refrigerant that is discharged from the compressor **11**, that is, discharge temperature, are arranged. In the vicinity of a refrigerant inlet of the accumulator **17** in the outdoor refrigerant pipe **26**, a suction pressure sensor **33** that detects suction pressure as pressure of the refrigerant that is sucked into the compressor **11**, and a suction temperature sensor **34** that detects temperature of the refrigerant that is sucked into the compressor **11** are arranged.

(30) In the outdoor liquid pipe **25** between the outdoor heat exchanger **13** and the outdoor unit expansion valve **14**, a refrigerant temperature sensor **35** that detects temperature of the refrigerant that flows into the outdoor heat exchanger **13** or temperature of the refrigerant that flows out of the outdoor heat exchanger **13** is arranged. Furthermore, in the vicinity of a suction port (not illustrated) of the outdoor unit **2**, an outdoor air temperature sensor **36** that detects temperature of outdoor air that flows into the outdoor unit **2**, that is, outdoor air temperature, is arranged.

(31) The control circuit **19** controls the entire air conditioner **1**. FIG. **3** is a block diagram illustrating an example of the control circuit **19** of the outdoor unit **2**. The control circuit **19** includes an acquisition unit **41**, a communication unit **42**, a storage unit **43**, a control unit **44**, a refrigerant amount estimation unit **45**, and an abnormality estimation unit **46**. The acquisition unit **41** acquires sensor values of detection units that are the various kinds of sensors as described above. The communication unit **42** is a communication interface for performing communication with a communication unit of each of the indoor units **3**. The storage unit **43** is, for example, a flash memory, and stores therein a control program of the outdoor unit **2**, operating state quantities, such as detection values, corresponding to detection signals from the various kinds of sensors, driving states of the compressor **11** and the outdoor unit fan **18**, operating information transmitted from each of the indoor units **3** (for example, including operating and stop information, an operating mode, such as cooling or heating, or the like), rated capacity of the outdoor unit **2**, requested capacity of each of the indoor units **3**, or the like. Further, the storage unit **43** includes an abnormality log storage unit **43A** that stores therein an abnormality log (to be described later).

(32) The control unit **44** periodically (for example, every 30 seconds) acquires the detected values that are obtained by the various kinds of sensors via the communication unit **42**, and receives input of signals including the operating state quantity that is transmitted from each of the indoor units **3** via the communication unit **42**. The control unit **44** adjusts the degree of opening of the outdoor unit expansion valve **14** and controls drive of the compressor **11** based on the various kinds of input information as described above.

(33) The refrigerant amount estimation unit **45** includes a refrigerant amount estimation model **45A** that estimates a refrigerant shortage rate of the refrigerant circuit **6** by using a detected value of a first feature value, where the first feature value represents an operating state quantity that is related to a refrigerant amount of the refrigerant circuit **6**. In the present embodiment, for example, a relative refrigerant amount is used as an amount of refrigerant that remains in the refrigerant circuit **6**. Specifically, the refrigerant amount estimation model **45A** is a model that estimates the refrigerant shortage rate of the refrigerant circuit **6** (indicating an amount of decrease from a

prescribed amount, where 100% indicates that the prescribed amount of refrigerant is stored, and the same applies to the following). The refrigerant amount estimation model **45A** includes a first cooler estimation model **45A1**, a second cooler estimation model **45A2**, a third cooler estimation model **45A3**, a first heater estimation model **45A4**, a second heater estimation model **45A5**, and a third heater estimation model **45A6**. Each of the refrigerant amount estimation models **45A** will be described in detail later.

(34) FIG. 4 is an explanatory diagram illustrating an example of pairing such that the outdoor unit **2** and each of the indoor units **3** are combined as a single pair. Meanwhile, for convenience of explanation, a case will be described in which the air conditioner **1** includes, for example, the single outdoor unit **2** and includes, for example, the four indoor units **3** (**3A**, **3B**, **3C**, **3D**) that are connected to the outdoor unit **2**. In this example, the single outdoor unit **2** and the single indoor unit **3** forms a single pair such that the outdoor unit **2** and the indoor unit **3A** forms a pair **P1**, the outdoor unit **2** and the indoor unit **3B** forms a pair **P2**, the outdoor unit **2** and the indoor unit **3C** forms a pair **P3**, and the outdoor unit **2** and the indoor unit **3D** forms a pair **P4**.

(35) The abnormality estimation unit **46** includes an abnormality estimation model **46A** that estimates whether the refrigerant circuit **6** is abnormal or normal for each of the pairs **P1** to **P4** of the outdoor unit **2** and the indoor units **3** by using a detected value of a second feature value that is related to abnormality of the refrigerant circuit **6** among the operating state quantities. The abnormality estimation unit **46** estimates abnormality of the refrigerant circuit **6** for each of the pairs **P1** to **P4**. If it is estimated that abnormality of the refrigerant circuit **6** has occurred in any of the pairs, it is estimated that the abnormality has occurred due to the indoor unit **3** of the subject pair. Further, if the abnormality estimation unit **46** estimates that abnormality has occurred in all of the pairs **P1** to **P4**, the abnormality estimation unit **46** estimates that the abnormality has occurred due to the outdoor unit **2**.

(36) The abnormality estimation model **46A** includes a cooling-period abnormality estimation model **46B** that is used when the air conditioner **1** performs cooling operation, and a heating-period abnormality estimation model **46C** that is used when the air conditioner **1** performs heating operation. Further, the abnormality estimation unit **46** includes a determination unit **46D** that is able to identify the outdoor unit **2** or the indoor unit **3** as a cause of the abnormality of the refrigerant circuit **6** based on an abnormality estimation result of each of the pairs **P1** to **P4**. Each of the abnormality estimation models **46A** as described above will be described in detail later.

(37) Configuration of Indoor Unit

(38) As illustrated in FIG. 2, the indoor unit **3** includes an indoor heat exchanger **51**, an indoor unit expansion valve **52**, a liquid pipe connection portion **53**, a gas pipe connection portion **54**, and an indoor unit fan **55**. The indoor heat exchanger **51**, the indoor unit expansion valve **52**, the liquid pipe connection portion **53**, and the gas pipe connection portion **54** are connected to one another by each of refrigerant pipes to be described later, and constitutes an indoor unit refrigerant circuit that is a part of the refrigerant circuit **6**.

(39) The indoor heat exchanger **51** performs heat exchange between the refrigerant and indoor air that is taken into the indoor unit **3** from a suction port (not illustrated) by rotation of the indoor unit fan **55**. One refrigerant port of the indoor heat exchanger **51** is connected to the liquid pipe connection portion **53** by an indoor liquid pipe **56**. Further, another refrigerant port of the indoor heat exchanger **51** is connected to the gas pipe connection portion **54** by an indoor gas pipe **57**. The indoor heat exchanger **51** functions as a condenser when the air conditioner **1** performs heating operation. In contrast, the indoor heat exchanger **51** functions as an evaporator when the air conditioner **1** performs cooling operation.

(40) The indoor unit expansion valve **52** is arranged in the indoor liquid pipe **56** and is an electronic expansion valve. When the indoor heat exchanger **51** functions as an evaporator, that is, when the indoor unit **3** performs cooling operation, a degree of opening of the indoor unit expansion valve **52** is adjusted such that a degree of superheat of refrigerant at a refrigerant outlet side (at the side of

the gas pipe connection portion **54**) of the indoor heat exchanger **51** reaches a target degree of superheat of the refrigerant. Further, when the indoor heat exchanger **51** functions as a condenser, that is, when the indoor unit **3** performs heating operation, the degree of opening of the indoor unit expansion valve **52** is adjusted such that a degree of supercooling of refrigerant at a refrigerant outlet side (at the side of the liquid pipe connection portion **53**) of the indoor heat exchanger **51** reaches the target degree of supercooling of the refrigerant. Here, the target degree of superheat of the refrigerant and the target degree of supercooling of the refrigerant are a degree of superheat of the refrigerant and a degree of supercooling of the refrigerant that are needed to cause the indoor unit **3** to fully demonstrate cooling capacity and heating capacity.

(41) The indoor unit fan **55** is made of a resin material and arranged in the vicinity of the indoor heat exchanger **51**. The indoor unit fan **55**, by being rotated by a fan motor (not illustrated), takes indoor air into the indoor unit **3** from a suction port (not illustrated), and discharges the indoor air that has been subjected to heat exchange with the refrigerant in the indoor heat exchanger **51** from a discharge port (not illustrated).

(42) Various sensors are arranged in the indoor unit **3**. In the indoor liquid pipe **56**, a liquid-side refrigerant temperature sensor **61** that detects temperature of the refrigerant that flows into the indoor heat exchanger **51** (heat exchange inlet temperature at the side of the indoor unit at the time of cooling operation) or temperature of the refrigerant that flows out of the indoor heat exchanger **51** (heat exchange outlet temperature at the side of the indoor unit at the time of heating operation) is arranged between the indoor heat exchanger **51** and the indoor unit expansion valve **52**. In the indoor gas pipe **57**, a gas-side temperature sensor **62** that detects temperature of the refrigerant that flows out of the indoor heat exchanger **51** (heat exchange outlet temperature at the side of the indoor unit at the time of cooling operation) or temperature of the refrigerant that flows into the indoor heat exchanger **51** (heat exchange inlet temperature at the side of the indoor unit at the time of heating operation) is arranged. In the vicinity of a suction port (not illustrated) of the indoor unit **3**, a suction temperature sensor **63** that detects temperature of the indoor air that flows into the indoor unit **3**, that is, suction temperature, is arranged.

(43) Operation of Refrigerant Circuit

(44) Flow of the refrigerant in the refrigerant circuit **6** and operation of each of the units when the air conditioner **1** of the present embodiment performs air conditioning operation will be described below. Meanwhile, arrows in FIG. **1** indicate flows of the refrigerant at the time of heating operation.

(45) When the air conditioner **1** performs heating operation, the four-way valve **12** is switched such that the first port **12A** and the fourth port **12D** communicate with each other and the second port **12B** and the third port **12C** communicate with each other. Accordingly, the refrigerant circuit **6** enters a heating cycle in which each of the indoor heat exchangers **51** functions as a condenser and the outdoor heat exchanger **13** functions as an evaporator. Meanwhile, for convenience of explanation, the flow of the refrigerant at the time of heating operation is indicated by bold arrows in FIG. **2**.

(46) If the compressor **11** drives when the refrigerant circuit **6** is in the state as described above, the refrigerant that is discharged from the compressor **11** flows through the discharge pipe **21**, flows into the four-way valve **12**, flows through the outdoor gas pipe **24** from the four-way valve **12**, and flows into the gas pipe **5** via the second stop valve **16**. The refrigerant that flows through the gas pipe **5** flows into each of the indoor units **3** in a distributed manner via each of the gas pipe connection portions **54**. The refrigerant that has flown into each of the indoor units **3** in a distributed manner flows through each of the indoor gas pipes **57** and flows into each of the indoor heat exchangers **51**. The refrigerant that has flown into each of the indoor heat exchangers **51** is subjected to heat exchange with the indoor air that is taken into each of the indoor units **3** by rotation of each of the indoor unit fans **55**, and condenses. In other words, each of the indoor heat exchangers **51** functions as a condenser and the indoor air that is heated by the refrigerant in each

of the indoor heat exchangers **51** is blown into a room from a discharge port (not illustrated), so that the room in which each of the indoor units **3** is installed is heated.

(47) The refrigerant that has flown into each of the indoor liquid pipes **56** from each of the indoor heat exchangers **51** is depressurized by passing through each of the indoor unit expansion valves **52** for which the degree of opening is adjusted such that the degree of supercooling of the refrigerant at a refrigerant outlet side of each of the indoor heat exchangers **51** reaches a target degree of supercooling of the refrigerant. Here, the target degree of supercooling of the refrigerant is determined based on cooling capacity that is needed in each of the indoor units **3**.

(48) The refrigerant that has been depressurized by each of the indoor unit expansion valves **52** flows out to the liquid pipe **4** from each of the indoor liquid pipes **56** via each of the liquid pipe connection portions **53**. The refrigerants that are collected in the liquid pipe **4** flow into the outdoor unit **2** via the first stop valve **15**. The refrigerant that has flown into the first stop valve **15** of the outdoor unit **2** flows through the outdoor liquid pipe **25** and depressurized by passing through the outdoor unit expansion valve **14**. The refrigerant that has been depressurized by the outdoor unit expansion valve **14** flows through the outdoor liquid pipe **25**, flows into the outdoor heat exchanger **13**, is subjected to heat exchange with the outdoor air that has flown through the suction port (not illustrated) of the outdoor unit **2** by rotation of the outdoor unit fan **18**, and evaporates. The refrigerant that has flown out to the outdoor refrigerant pipe **26** from the outdoor heat exchanger **13** flows into the four-way valve **12**, the outdoor refrigerant pipe **26**, the accumulator **17**, and the suction pipe **22** in this order, is sucked into the compressor **11** where the refrigerant is compressed again, and flows out to the outdoor gas pipe **24** by the first port **12A** and the fourth port **12D** of the four-way valve **12**.

(49) Further, when the air conditioner **1** performs cooling operation, the four-way valve **12** is switched such that the first port **12A** and the second port **12B** communicate with each other and the third port **12C** and the fourth port **12D** communicate with each other. Accordingly, the refrigerant circuit **6** enters a cooling cycle in which each of the indoor heat exchangers **51** functions as an evaporator and the outdoor heat exchanger **13** functions as a condenser. Meanwhile, for convenience of explanation, the flow of the refrigerant at the time of cooling operation is indicated by dashed arrows in FIG. 2.

(50) If the compressor **11** drives when the refrigerant circuit **6** is in the state as described above, the refrigerant that is discharged from the compressor **11** flows through the discharge pipe **21**, flows into the four-way valve **12**, flows through the outdoor refrigerant pipe **26** from the four-way valve **12**, and flows into the outdoor heat exchanger **13**. The refrigerant that has flown into the outdoor heat exchanger **13** is subjected to heat exchange with outdoor air that is taken into the outdoor unit **2** by rotation of the outdoor unit fan **18**, and condenses. In other words, the outdoor heat exchanger **13** functions as a condenser and the indoor air that is heated by the refrigerant in the outdoor heat exchanger **13** is blown out of the room from a discharge port (not illustrated).

(51) The refrigerant that has flown into the outdoor liquid pipe **25** from the outdoor heat exchanger **13** is depressurized by passing through the outdoor unit expansion valve **14** for which the degree of opening is adjusted to full-open. The refrigerant that has been depressurized by the outdoor unit expansion valve **14** flows through the liquid pipe **4** via the first stop valve **15** and flows into each of the indoor units **3** in a distributed manner. The refrigerant that has flown into each of the indoor units **3** in a distributed manner flows through the indoor liquid pipe **56** via each of the liquid pipe connection portions **53** and is depressurized by passing through the indoor unit expansion valve **52** for which the degree of opening is adjusted such that the degree of supercooling of the refrigerant at the refrigerant outlet of the indoor heat exchanger **51** reaches the target degree of supercooling of the refrigerant. The refrigerant that has been depressurized by the indoor unit expansion valve **52** flows through the indoor liquid pipe **56**, flows into the indoor heat exchanger **51**, is subjected to heat exchange with the indoor air that has flown in from the suction port (not illustrated) of the indoor unit **3** by rotation of the indoor unit fan **55**, and evaporates. In other words, each of the

indoor heat exchangers **51** functions as an evaporator and the indoor air that is cooled by the refrigerant in each of the indoor heat exchangers **51** is blown into the room from a discharge port (not illustrated), so that the room in which each of the indoor units **3** is installed is cooled.

(52) The refrigerant that flows into the gas pipe **5** from the indoor heat exchanger **51** via the gas pipe connection portion **54** flows through the outdoor gas pipe **24** via the second stop valve **16** of the outdoor unit **2**, and flows into the fourth port **12D** of the four-way valve **12**. The refrigerant that has flown into the fourth port **12D** of the four-way valve **12** flows into the refrigerant inflow side of the accumulator **17** via the third port **12C**. The refrigerant that has flown in from the refrigerant inflow side of the accumulator **17** flows in via the suction pipe **22**, is sucked by the compressor **11**, and is compressed again.

(53) The acquisition unit **41** in the control circuit **19** acquires sensor values of the discharge pressure sensor **31**, the discharge temperature sensor **32**, the suction pressure sensor **33**, the suction temperature sensor **63**, the refrigerant temperature sensor **35**, and the outdoor air temperature sensor **36** in the outdoor unit **2**. Further, the acquisition unit **41** acquires sensor values of the liquid-side refrigerant temperature sensor **61**, the gas-side temperature sensor **62**, and the suction temperature sensor **63** of each of the indoor units **3**.

(54) FIG. **5** is a Mollier diagram illustrating a cooling cycle of the air conditioner **1**. When the air conditioner **1** performs cooling operation, the outdoor heat exchanger **13** functions as a condenser and the indoor heat exchanger **51** functions as an evaporator. Further, when the air conditioner **1** performs heating operation, the outdoor heat exchanger **13** functions as an evaporator and the indoor heat exchanger **51** functions as a condenser.

(55) The compressor **11** compresses a low-temperature and low-pressure gas refrigerant that flows in from the evaporator, and discharges a high-temperature and high-pressure gas refrigerant (a refrigerant in the state at a point B in FIG. **5**). Meanwhile, the temperature of the gas refrigerant that is discharged from the compressor **11** is discharge temperature and the discharge temperature is detected by the discharge temperature sensor **32**.

(56) The condenser performs heat exchange between the high-temperature and high-pressure gas refrigerant coming from the compressor **11** with air, and condenses the gas refrigerant. In this case, in the condenser, the entire gas refrigerant turns into a liquid refrigerant due to a latent heat change, and thereafter, the temperature of the liquid refrigerant is reduced due to a sensible heat change, so that a supercooled state is achieved (a state at a point C in FIG. **5**). Meanwhile, the temperature at which the gas refrigerant is changed to the liquid refrigerant due to the latent heat change is high-pressure saturation temperature, and the temperature of the refrigerant in the supercooled state at an outlet of the condenser is the heat exchange outlet temperature. The high-pressure saturation temperature is temperature that corresponds to a pressure value (a pressure value P2 that is represented by “HPS” in FIG. **5**) that is detected by the discharge pressure sensor **31**. The heat exchange outlet temperature is the temperature of the refrigerant that flows through the outdoor liquid pipe **25** and is detected by the refrigerant temperature sensor **35**.

(57) The expansion valve depressurizes the low-temperature and high-pressure refrigerant that has flown out of the condenser, so that a gas-liquid two-phase refrigerant in which gas and liquid are mixed is obtained (a refrigerant in a state at a point D in FIG. **5**).

(58) The evaporator performs heat exchange between the gas-liquid two-phase refrigerant that has flown in and air, and evaporates the refrigerant. In this case, in the evaporator, after the entire gas-liquid two-phase refrigerant turns into a gas refrigerant due to a latent heat change, temperature of the gas refrigerant increases due to a sensible heat change, so that the refrigerant enters in a superheated state (a state at a point A in FIG. **5**) and is sucked into the compressor **11**. Meanwhile, the temperature at which the liquid refrigerant is changed to the gas refrigerant due to the latent heat change is low-pressure saturation temperature. The low-pressure saturation temperature is temperature that corresponds to a pressure value (a pressure value P1 indicated by “LPS” in FIG. **5**) that is detected by the suction pressure sensor **33**. Further, the temperature of the refrigerant that is

superheated by the evaporator and sucked into the compressor **11** is suction temperature. The suction temperature is detected by the suction temperature sensor **34**.

(59) Meanwhile, the degree of supercooling of the refrigerant that is in the supercooled state when the refrigerant flows out of the condenser may be calculated by subtracting the temperature (the heat exchange outlet temperature as described above) of the refrigerant at a refrigerant outlet of the heat exchanger that functions as a condenser from the high-pressure saturation temperature. Furthermore, the degree of suction superheat of the refrigerant that is in the superheated state when the refrigerant flows out of the evaporator may be calculated by subtracting the suction temperature from the low-pressure saturation temperature.

(60) First Feature Value

(61) FIG. **6** is an explanatory diagram illustrating examples of a first feature value that is used for the first to the third cooler estimation models **45A1**, **45A2**, and **45A3** and a second feature value that is used for the cooling-period abnormality estimation model **46B**. The first feature value is adopted as an operating state quantity that is used for the refrigerant amount estimation model **45A**. Examples of the first feature value that is used for the first to the third cooler estimation models **45A1**, **45A2**, and **45A3** include a rotation speed of the compressor **11**, the high-pressure saturation temperature, the suction temperature, low-pressure refrigerant temperature, the degree of supercooling of refrigerant (outdoor heat exchange subcool), and the outdoor air temperature. The rotation speed of the compressor **11** is detected by a rotation speed sensor (not illustrated) of the compressor **11**. The high-pressure saturation temperature is a value that is obtained by converting the pressure value detected by the discharge pressure sensor **31** to temperature. The suction temperature is detected by the suction temperature sensor **34**. The low-pressure refrigerant temperature is temperature of the refrigerant that is superheated by the evaporator and sucked into the compressor **11**. The degree of supercooling of the refrigerant is, for example, a value that is calculated by subtracting the outdoor heat exchange outlet temperature from the high-pressure saturation temperature. The outdoor air temperature is detected by the outdoor air temperature sensor **36**. Meanwhile, the outdoor heat exchange outlet temperature is detected by the refrigerant temperature sensor **35**. For example, the operating state quantity including the first feature value used for the first to the third cooler estimation models **45A1**, **45A2** and **45A3** is periodically detected by a detection unit, such as the rotation speed sensor, the discharge pressure sensor **31**, the suction temperature sensor **34**, the outdoor air temperature sensor **36**, or the refrigerant temperature sensor **35**. Meanwhile, if the air conditioner **1** is operating, the control unit **44** instructs the detection unit to periodically (for example, every 10 minutes) acquires the operating state quantity. The detection unit that has received the instruction detects the operating state quantity from the various kinds of sensors that are arranged in the air conditioner **1**. Acquisition time information is also given to the operating state quantity that is periodically acquired.

(62) FIG. **7** is an explanatory diagram illustrating examples of the first feature value that is used for the first to the third heater estimation models **45A4**, **45A5**, and **45A6** and the second feature value that is used for the heating-period abnormality estimation model **46C**. Examples of the first feature value used for the first to the third heater estimation models **45A4**, **45A5**, and **45A6** include the degree of opening of the outdoor unit expansion valve **14**, the rotation speed of the compressor **11**, the degree of suction superheat, and the outdoor air temperature. The degree of opening of the outdoor unit expansion valve **14** is the number of pulses that the control unit **44** gives to a stepping motor (not illustrated) of the outdoor unit expansion valve **14**. The rotation speed of the compressor **11** is detected by the rotation speed sensor (not illustrated) of the compressor **11**. The degree of suction superheat is a value that is calculated by, for example, subtracting the low-pressure saturation temperature from the suction temperature. The outdoor air temperature is detected by the outdoor air temperature sensor **36**. The suction temperature is detected by the suction temperature sensor **34**, and the low-pressure saturation temperature is a value that is obtained by converting the pressure value detected by the suction pressure sensor **33** to temperature. Meanwhile, for example,

the operating state quantity including the first feature value that is used for the first to the third heater estimation models **45A4**, **45A5** and **45A6** is periodically detected by the detection unit, such as the rotation speed sensor, the suction temperature sensor **34**, or the outdoor air temperature sensor **36**.

(63) Second Feature Value

(64) The operating state quantity that is used for the abnormality estimation model **46A** includes the second feature value that is related to abnormality of the refrigerant circuit **6**. The second feature value that is used to generate the abnormality estimation model **46A** is a value that is obtained when, for example, the refrigerant circuit **6** is realized on the computer, numerical analysis is performed (hereinafter, performance of numerical analysis may be described as a simulation), operation of the refrigerant circuit **6** is normal, and only a remaining refrigerant amount is changed. Meanwhile, the second feature value that is used to generate the abnormality estimation model **46A** will be referred to as a simulation value (may be simply referred to as a “value”). The second feature value includes at least a single operating state quantity that is included in the first feature value and a single operating state quantity that is not included in the first feature value.

(65) Examples of the second feature value that is used for the cooling-period abnormality estimation model **46B** include, as illustrated in FIG. **6**, the rotation speed of the compressor **11**, the high-pressure saturation temperature, the suction temperature, the low-pressure refrigerant temperature, the outdoor air temperature, a high-pressure sensor (HPS), and the heat exchange outlet temperature. The rotation speed of the compressor **11** is detected by the rotation speed sensor (not illustrated) of the compressor **11**. The high-pressure saturation temperature is a value that is obtained by converting the pressure value detected by the discharge pressure sensor **31** to temperature. The suction temperature is detected by the suction temperature sensor **34**. The low-pressure refrigerant temperature is temperature of the refrigerant that is superheated by the evaporator and sucked into the compressor **11**. The outdoor air temperature is detected by the outdoor air temperature sensor **36**. The high-pressure sensor is a pressure value that is detected by the discharge pressure sensor **31**. The heat exchange outlet temperature is detected by the refrigerant temperature sensor **35**. Meanwhile, for example, the operating state quantity including the second feature value that is used for the cooling-period abnormality estimation model **46B** is periodically detected by the detection unit, such as the rotation speed sensor, the discharge pressure sensor **31**, the suction temperature sensor **34**, the outdoor air temperature sensor **36**, or the refrigerant temperature sensor **35**.

(66) Further, examples of the second feature value that is used for the heating-period abnormality estimation model **46C** include, as illustrated in FIG. **7**, the degree of opening of the outdoor unit expansion valve **14**, the rotation speed of the compressor **11**, the outdoor air temperature, the discharge temperature, the suction temperature, the low-pressure saturation temperature, and a low-pressure sensor (LPS). The degree of opening of the outdoor unit expansion valve **14** is detected by a sensor (not illustrated). The rotation speed of the compressor **11** is detected by the rotation speed sensor (not illustrated) of the compressor **11**. The outdoor air temperature is detected by the outdoor air temperature sensor **36**. The discharge temperature is detected by the discharge temperature sensor **32**. The suction temperature is detected by the suction temperature sensor **34**. The low-pressure saturation temperature is a value that is obtained by converting the pressure value detected by the suction pressure sensor **33** to temperature. The low-pressure sensor is the pressure value that is detected by the suction pressure sensor **33**. Meanwhile, for example, the operating state quantity including the second feature value that is used for the heating-period abnormality estimation model **46C** is periodically detected by the detection unit, such as the rotation speed sensor, the suction temperature sensor **34**, the outdoor air temperature sensor **36**, or the suction pressure sensor **33**.

(67) The second feature value that is commonly used between the cooling-period abnormality estimation model **46B** and the heating-period abnormality estimation model **46C** includes the rotation speed of the compressor **11** and the suction temperature that are the operating state

quantities at the side of the outdoor unit **2**.

(68) Further, the second feature value that is commonly used between the cooling-period abnormality estimation model **46B** and the heating-period abnormality estimation model **46C** includes the operating state quantity at the side of the indoor unit **3**; for example, heat exchange inlet temperature at the side of the indoor unit (detected by the liquid-side refrigerant temperature sensor **61** and detected by the gas-side temperature sensor **62** at the time of heating operation), the heat exchange outlet temperature at the side of the indoor unit (detected by the gas-side temperature sensor **62** at the time of cooling operation and detected by the liquid-side refrigerant temperature sensor **61** at the time of heating operation), and the degree of opening of the indoor unit expansion valve **52**. Meanwhile, as the second feature value at the side of the indoor unit **3**, for example, the heat exchange inlet temperature at the side of the indoor unit, the heat exchange outlet temperature at the side of the indoor unit, and the degree of opening of the indoor unit expansion valve **52** are illustrated, but it is possible to acquire the feature value in the same manner even if the indoor unit **3** is of a different type, such as a duct type or a ceiling cassette type.

(69) Configuration of Refrigerant Amount Estimation Model

(70) The refrigerant amount estimation model **45A** is generated by using a detected value of the first feature value. The refrigerant amount estimation unit **45** estimates a refrigerant shortage rate of the refrigerant circuit **6** by applying the detected value of the first feature value, which is acquired at a timing different from a timing of generation of the refrigerant amount estimation model **45A**, to the refrigerant amount estimation model **45A**.

(71) The refrigerant amount estimation model **45A** is generated by a multiple regression analysis method that is one of regression analysis methods by using an arbitrary operating state quantity (the detected value of the first feature value) among the plurality of operating state quantities. In the multiple regression analysis method, the refrigerant amount estimation model **45A** is generated by selecting a regression equation, in which a P value (a value that indicates a degree of influence of the operating state quantity on accuracy of the generated estimation model (predetermined weight parameter)) is minimized and a correction value R2 (a value that indicates accuracy of the generated refrigerant amount estimation model **45A**) is maximized in a range from 0.9 to 1.0, from among regression equations that are obtained from a plurality of simulation results (results obtained by reproducing the refrigerant circuit **6** by a numerical calculation and calculating values of the operating state quantities with respect to the remaining refrigerant amount). Here, the P value and the correction value R2 are values that are related to accuracy of the refrigerant amount estimation model **45A** when the refrigerant amount estimation model **45A** is generated by the multiple regression analysis method, and the accuracy of the generated refrigerant amount estimation model **45A** increases as the P value decreases and as the correction value R2 approaches 1.0. As a result, if the refrigerant shortage rate is 0% to 30% at the time of cooling, for example, the operating state quantities, such as the degree of supercooling of the refrigerant, the outdoor air temperature, the high-pressure saturation temperature, and the rotation speed of the compressor **11**, are adopted as the first feature values. If the refrigerant shortage rate is 40% to 70% at the time of cooling, for example, the operating state quantities, such as the suction temperature, the outdoor air temperature, and the rotation speed of the compressor **11**, are adopted as the first feature values. If the refrigerant shortage rate at the time of heating is 0% to 20%, for example, the operating state quantity, such as the degree of opening of the outdoor unit expansion valve **14**, is adopted as the feature value. Further, if the refrigerant shortage rate at the time of heating is 30% to 70%, for example, the operating state quantities, such as the degree of suction superheat (the suction temperature-the low-pressure saturation temperature), the outdoor air temperature, the rotation speed of the compressor **11**, and the degree of opening of the outdoor unit expansion valve **14**, are adopted as the first feature values.

(72) The refrigerant amount estimation model **45A** includes the first cooler estimation model **45A1**, the second cooler estimation model **45A2**, the third cooler estimation model **45A3**, the first heater

estimation model **45A4**, the second heater estimation model **45A5**, and the third heater estimation model **45A6** as described above. In the present embodiment, each of the estimation models as described above is generated by using a simulation result to be described later, and is stored in the refrigerant amount estimation unit **45** in the control circuit **19** of the air conditioner **1** in advance.

(73) The first cooler estimation model **45A1** is the refrigerant amount estimation model **45A** that is effective when the refrigerant shortage rate is 0% to 30% (first range), and is a first regression equation that is able to estimate the refrigerant shortage rate with high accuracy. The first regression equation is, for example, $(\alpha 1 \times \text{the degree of supercooling of the refrigerant}) + (\alpha 2 \times \text{the outdoor air temperature}) + (\alpha 3 \times \text{the high-pressure saturation temperature}) + (\alpha 4 \times \text{the rotation speed of the compressor } \mathbf{11}) + \alpha 5$. It is assumed that the coefficients $\alpha 1$ to $\alpha 5$ are determined when the estimation models are generated. The refrigerant amount estimation unit **45** calculates the refrigerant shortage rate of the refrigerant circuit **6** at a current time by assigning the degree of supercooling of the refrigerant, the outdoor air temperature, the high-pressure saturation temperature, and the rotation speed of the compressor **11** at the current time, which are acquired by the acquisition unit **41**, to the first regression equation. Meanwhile, the reason that the degree of supercooling of the refrigerant, the outdoor air temperature, the high-pressure saturation temperature, and the rotation speed of the compressor **11** are assigned is to use the first feature values that are used to generate the first cooler estimation model **45A1**. The degree of supercooling of the refrigerant can be calculated by, for example, subtracting the heat exchange outlet temperature from the high-pressure saturation temperature. The outdoor air temperature is detected by the outdoor air temperature sensor **36**. The high-pressure saturation temperature is a value that is obtained by converting the pressure value detected by the discharge pressure sensor **31** to temperature. The rotation speed of the compressor **11** is detected by the rotation speed sensor (not illustrated) of the compressor **11**.

(74) The second cooler estimation model **45A2** is the refrigerant amount estimation model **45A** that is effective when the refrigerant shortage rate is 40% to 70% (second range), and is a second regression equation that is able to estimate the refrigerant shortage rate with high accuracy. The second regression equation is, for example, $(\alpha 11 \times \text{the suction temperature}) + (\alpha 12 \times \text{the outdoor air temperature}) + (\alpha 13 \times \text{the rotation speed of the compressor } \mathbf{11}) + \alpha 14$. It is assumed that the coefficients $\alpha 11$ to $\alpha 14$ are determined when the estimation model is generated. The refrigerant amount estimation unit **45** calculates the refrigerant shortage rate of the refrigerant circuit **6** at a current time by assigning the suction temperature, the outdoor air temperature, and the rotation speed of the compressor **11** at the current time, which are acquired by the acquisition unit **41**, to the second regression equation. Meanwhile, the reason that the suction temperature, the outdoor air temperature, and the rotation speed of the compressor **11** are assigned is to use the feature values that are used to generate the second cooler estimation model **45A2**. The suction temperature is detected by the suction temperature sensor **34**. The outdoor air temperature is detected by the outdoor air temperature sensor **36**. The rotation speed of the compressor **11** is detected by the rotation speed sensor (not illustrated) of the compressor **11**.

(75) Meanwhile, as described above, the refrigerant shortage rate that can be obtained by the first regression equation is 0% to 30%, and the refrigerant shortage rate that can be obtained by the second regression equation is 40% to 70%. In this case, when the refrigerant shortage rate is 30% to 40%, and if the first regression equation is used, the refrigerant shortage rate is calculated as 30%, whereas if the second regression equation is used, the refrigerant shortage rate is calculated as 40%. In other words, if the refrigerant shortage rate is 30% to 40%, both of the degree of supercooling of the refrigerant, which is highly contributable when the refrigerant shortage rate is equal to or smaller than 30%, and the suction temperature, which is highly contributable when the refrigerant shortage rate is equal to or larger than 40%, are less likely to change, so that it is difficult to generate an effective estimation model. Therefore, if the first regression equation or the second regression equation is used, the refrigerant shortage rate largely differs depending on the

model to be used as illustrated in FIG. 8A.

(76) The third cooler estimation model **45A3** is a cooling-period refrigerant shortage rate calculation formula that can cover the refrigerant shortage rate in a range of 0% to 70% that includes a range in which it is difficult to estimate the refrigerant shortage rate by using any of the first regression equation and the second regression equation as described above. As illustrated in FIG. 8B, the cooling-period refrigerant shortage rate calculation formula continuously connects a refrigerant shortage rate that is an estimation result obtained by the first regression equation and a refrigerant shortage rate that is an estimation result obtained by the second regression equation, by a sigmoid curve using a sigmoid coefficient. Specifically, the cooling-period refrigerant shortage rate calculation formula is (the sigmoid coefficient×the refrigerant shortage rate obtained by the first regression equation)+((1−the sigmoid coefficient)×the refrigerant shortage rate obtained by the second regression equation). The refrigerant amount estimation unit **45** calculates the refrigerant shortage rate of the refrigerant circuit **6** at a current time by assigning each of the refrigerant shortage rates, which are calculated by assigning the current operating state quantities that are acquired by the acquisition unit **41** to the first regression equation and the second regression equation, to the cooling-period refrigerant shortage rate calculation formula.

(77) Here, the sigmoid coefficient is calculated by using any of the operating state quantities. In the present embodiment, by taking into account the fact that a result obtained by the first regression equation becomes approximately constant if the subcool reaches 0, a calculation formula is determined such that the sigmoid coefficient is 0.5 when the subcool is 5° C.

$p=1/(1+\exp(-(sc-5)))$ p: sigmoid coefficient sc: subcool value

(78) If the sigmoid coefficient is determined as described above and the sigmoid coefficient is used for the third cooler estimation model **45A3**, the estimated value of the first cooler estimation model **45A1** is dominant in the estimated value obtained by the third cooler estimation model **45A3** when the refrigerant shortage rate is 0% to 30%, that is, when the refrigerant shortage rate is in the first range, and, the estimated value of the second cooler estimation model **45A2** is dominant in the estimated value obtained by the third cooler estimation model **45A3** when the refrigerant shortage rate is 40% to 70%, that is, when the refrigerant shortage rate is in the second range.

(79) Meanwhile, the sigmoid coefficient need not always be calculated by the method as described above, but it is sufficient to determine the sigmoid coefficient such that when an actual refrigerant shortage rate is equal to or larger than 30%, that is, when the actual refrigerant shortage rate does not fall in the first range, the estimated value of the second cooler estimation model **45A2** becomes dominant in the estimated value obtained by the third cooler estimation model **45A3**, and when the actual refrigerant shortage rate is equal to or smaller than 40%, that is, when the actual refrigerant shortage rate does not fall in the second range, the estimated value of the first cooler estimation model **45A1** becomes dominant in the estimated value obtained by the third cooler estimation model **45A3**.

(80) The first heater estimation model **45A4** is the refrigerant amount estimation model **45A** that is effective when the refrigerant shortage rate is 0% to 20% (third range), and is a fourth regression equation that is able to estimate the refrigerant shortage rate with high accuracy. The fourth regression equation is, for example, ($\alpha 31 \times$ the degree of opening of the outdoor unit expansion valve **14**)+ $\alpha 32$. The refrigerant amount estimation unit **45** calculates the refrigerant shortage rate by assigning the current degree of opening of the outdoor unit expansion valve **14** acquired by the acquisition unit **41** to the fourth regression equation. Meanwhile, the reason that the degree of opening of the outdoor unit expansion valve **14** is assigned is to use the feature value that is used to generate the first heater estimation model **45A4**.

(81) The second heater estimation model **45A5** is the refrigerant amount estimation model **45A** that is effective when the refrigerant shortage rate is 30% to 70% (fourth range), and is a fifth regression equation that is able to estimate the refrigerant shortage rate with high accuracy. The fifth regression equation is, for example, ($\alpha 41 \times$ the degree of suction superheat)+($\alpha 42 \times$ the outdoor

air temperature)+(α43×the rotation speed of the compressor 11)+(α44×the degree of opening of the outdoor unit expansion valve 14)+α45. The coefficients α41 to α45 are determined when the estimation models are generated. The refrigerant amount estimation unit 45 calculates the refrigerant shortage rate of the refrigerant circuit 6 at a current time by assigning the degree of suction superheat, the outdoor air temperature, the rotation speed of the compressor 11, and the degree of opening of the expansion valve on the main side at the current time, which are acquired by the acquisition unit 41, to the fifth regression equation. Meanwhile, the reason that the degree of suction superheat, the outdoor air temperature, the rotation speed of the compressor 11, and the degree of opening of the outdoor unit expansion valve 14 are assigned is to use the feature values that are used to generate the second heater estimation model 45A5. The degree of suction superheat can be calculated by, for example, subtracting the low-pressure saturation temperature from the suction temperature. The outdoor air temperature is detected by the outdoor air temperature sensor 36. The rotation speed of the compressor 11 is detected by the rotation speed sensor (not illustrated) of the compressor 11. The degree of opening of the outdoor unit expansion valve 14 is calculated by a sensor (not illustrated).

(82) Further, as described above, the refrigerant shortage rate that can be obtained by the fourth regression equation is 0% to 20%, and the refrigerant shortage rate that can be obtained by the fifth regression equation is 30% to 70%. In this case, when the refrigerant shortage rate is 20% to 30%, and if the fourth regression equation is used, the refrigerant shortage rate is calculated as 20%, whereas if the fifth regression equation is used, the refrigerant shortage rate is calculated as 30%. In other words, if the refrigerant shortage rate is 20% to 30%, both of the degree of opening of the outdoor unit expansion valve 14, which is highly contributable when the refrigerant shortage rate is equal to or smaller than 20%, and the degree of suction superheat, which is highly contributable when the refrigerant shortage rate is equal to or larger than 30%, are less likely to change, so that it is difficult to generate an effective estimation model. Therefore, if the fourth regression equation or the fifth regression equation is used, the refrigerant shortage rate largely differs depending on the model to be used as illustrated in FIG. 9A.

(83) The third heater estimation model 45A6 is a heating-period refrigerant shortage rate calculation formula that can cover the refrigerant shortage rate in a range from 0% to 70% that includes a range in which it is difficult to estimate the refrigerant shortage rate by using any of the fourth regression equation and the fifth regression equation as described above. As illustrated in FIG. 9B, the heating-period refrigerant shortage rate calculation formula continuously connects a refrigerant shortage rate that is an estimation result obtained by the fourth regression equation and a refrigerant shortage rate that is an estimation result obtained by the fifth regression equation, by a sigmoid curve using a sigmoid coefficient. Specifically, the heating-period refrigerant shortage rate calculation formula is (the sigmoid coefficient×the refrigerant shortage rate obtained by the fifth regression equation)+(1−the sigmoid coefficient)×the refrigerant shortage rate obtained by the fourth regression equation). The refrigerant amount estimation unit 45 calculates the refrigerant shortage rate of the refrigerant circuit 6 at a current time by assigning each of the refrigerant shortage rates, which are calculated by assigning the current operating state quantities that are acquired by the acquisition unit 41 to the fourth regression equation and the fifth regression equation, to the heating-period refrigerant shortage rate calculation formula.

(84) Here, the sigmoid coefficient is calculated by using any of the operating state quantities, in the same manner as in the cooling operation. In the present embodiment, by taking into account the fact that a result obtained by the fourth regression equation becomes approximately constant if the degree of opening of the outdoor unit expansion valve 14 is set to full-open based on the assumption that a fully-closed state is indicated by 0 and a fully-opened state is indicated by 100, a calculation formula is determined such that the sigmoid coefficient is 0.5 when the degree of opening of the outdoor unit expansion valve 14 is 90.

$p=1/(1+\exp(-(D/10-45)))$ p: sigmoid coefficient D: degree of opening of the outdoor unit

expansion valve **14**

(85) If the sigmoid coefficient is determined as described above and the sigmoid coefficient is used for the third heater estimation model **45A6**, the estimated value of the first heater estimation model **45A4** is dominant in the estimated value obtained by the third heater estimation model **45A6** when the refrigerant shortage rate is 0% to 20%, that is, when the refrigerant shortage rate is in the third range, and, the estimated value of the second heater estimation model **45A5** is dominant in the estimated value obtained by the third heater estimation model **45A6** when the refrigerant shortage rate is 30% to 70%, that is, when the refrigerant shortage rate is in the fourth range.

(86) Meanwhile, the sigmoid coefficient need not always be calculated by the method as described above, but it is sufficient to determine the sigmoid coefficient such that when an actual refrigerant shortage rate is equal to or larger than 20%, that is, when the actual refrigerant shortage rate does not fall in the third range, the estimated value of the second heater estimation model **45A5** becomes dominant in the estimated value obtained by the third heater estimation model **45A6**, and when the actual refrigerant shortage rate is equal to or smaller than 30%, that is, when the actual refrigerant shortage rate does not fall in the fourth range, the estimated value of the first heater estimation model **45A4** becomes dominant in the estimated value obtained by the third heater estimation model **45A6**.

(87) As described above, the refrigerant shortage rate is estimated by using the first regression equation, the second regression equation, and the cooling-period refrigerant shortage rate calculation formula at the time of cooling operation. If a value of the degree of supercooling of the refrigerant at the time of cooling is larger than a first threshold (Tv1 in FIG. 8A and FIG. 8B), it is possible to estimate the refrigerant shortage rate with increased accuracy by selecting the first regression equation rather than selecting the second regression equation. Further, if the value of the degree of supercooling of the refrigerant at the time of cooling is smaller than the first threshold, it is possible to estimate the refrigerant shortage rate with increased accuracy by selecting the second regression equation rather than selecting the first regression equation. Furthermore, if the value of the degree of supercooling of the refrigerant at the time of cooling is around the first threshold, the estimated value of the refrigerant shortage rate largely differs depending on the regression equation to be used. Therefore, at the time of cooling, the cooling-period refrigerant shortage rate calculation formula that includes the first regression equation and the second regression equation is selected. With this configuration, it is possible to estimate the refrigerant shortage rate at the time of cooling with high accuracy.

(88) Moreover, the refrigerant shortage rate is estimated by using the fourth regression equation, the fifth regression equation, and the heating-period refrigerant shortage rate calculation formula at the time of heating operation. If the degree of opening of the outdoor unit expansion valve **14** at the time of heating is smaller than a second threshold (Tv2 in FIG. 9A and FIG. 9B), it is possible to estimate the refrigerant shortage rate with increased accuracy by selecting the fourth regression equation rather than selecting the fifth regression equation. Further, if the degree of opening of the outdoor unit expansion valve **14** at the time of heating is not smaller than the second threshold, it is possible to estimate the refrigerant shortage rate with increased accuracy by selecting the fifth regression equation rather than selecting the fourth regression equation. Furthermore, if a value of the degree of opening of the outdoor unit expansion valve **14** is around the second threshold, the estimated value of the refrigerant shortage rate largely differs depending on the regression equation to be used. Therefore, at the time of heating, the heating-period refrigerant shortage rate calculation formula that includes the fourth regression equation and the fifth regression equation is selected. With this configuration, it is possible to estimate the refrigerant shortage rate at the time of heating with high accuracy.

(89) Configuration of Abnormality Estimation Model

(90) The abnormality estimation model **46A** is generated by using a simulation value that is a value of the second feature value that is obtained as a result of a simulation of operation of the refrigerant

circuit when the refrigerant circuit **6** operates normally and when only the remaining refrigerant amount is changed. The abnormality estimation unit **46** estimates whether the detected value of the second feature value of each of the pairs **P1** to **P4** is abnormal or normal by applying the detected value of the second feature value of each of the pairs **P1** to **P4** acquired from the operating air conditioner **1** to the abnormality estimation model **46A**. Specifically, if the detected value of the second feature value of each of the pairs **P1** to **P4** is abnormal, the abnormality estimation unit **46** estimates that abnormality has occurred in the refrigerant circuit **6** of each of the pairs **P1** to **P4**. If the detected value of the second feature value of each of the pairs **P1** to **P4** is normal, the abnormality estimation unit **46** estimates that the refrigerant circuit **6** of each of the pairs **P1** to **P4** is normal.

(91) To generate the abnormality estimation model **46A**, for example, a Kernel density estimation method is adopted. The Kernel density estimation method is a method of estimating an entire distribution from limited sample points. The abnormality estimation model **46A** calculates a degree of deviation (hereinafter, may also be referred to as an outlier) from a local maximum value (a center of a cluster (a set of data having similarities)) of a density function, based on the density function of the entire distribution that is estimated from the limited sample points. Further, if determination target data is input, the abnormality estimation model **46A** calculates an outlier of the data and determines whether the outlier falls within a predetermined range or not (whether the determination target data is included in the cluster).

(92) FIG. **10** is an explanatory diagram illustrating an example of the way of distribution of detected values of the second feature values of the abnormality estimation model **46A**. As illustrated in FIG. **10**, the abnormality estimation model **46A** adopts, as a single cluster, a set of values of the second feature values (hereinafter, also referred to as “simulation values of the second feature values”) that are obtained by a simulation in a steady state and a refrigerant leaked state while the refrigerant circuit **6** is in a normal state, and classifies this cluster as normal. The detected values of the second feature values in the steady state are detected values of the second feature values that are obtained by a simulation of operation of the normal refrigerant circuit **6**. A condition for the simulation is the steady state while the refrigerant circuit **6** is in the normal state or the state in which a refrigerant storage amount is reduced (the refrigerant leaked state). The simulation values of the second feature values in the normal state are values of the second feature values that are obtained by a simulation that is performed by assuming a state in which each of the components (the refrigerant circuit **6**, the compressor, the expansion valve, and the like) included in the air conditioner **1** operates normally. Further, the simulation values of the second feature values in the refrigerant leaked state are values of the second feature values that are obtained by a simulation that is performed by assuming a state in which each of the components (the refrigerant circuit **6**, the compressor, the expansion valve, and the like) included in the air conditioner operates normally and by assuming a state in which only an amount of the refrigerant remaining in the refrigerant circuit **6** is changed (reduced). Furthermore, if a detected value of the second feature value that deviates from the cluster that is classified as normal by the abnormality estimation model **46A** is input, the detected value is classified as abnormal. Meanwhile, the detected value that is classified as abnormal is a detected value that deviates from the cluster that is classified as normal when the detected value is plotted on a graph as illustrated in FIG. **10**. Moreover, abnormal indicates a state in which a failure is highly likely to have occurred in a device included in the refrigerant circuit **6**.

(93) The abnormality estimation model **46A** quantifies a difference between the value of the second feature value that is obtained by the simulation in the steady state and the refrigerant leaked state while the refrigerant circuit **6** is in the normal state and the detected value of the second feature value of each of the pairs **P1** to **P4** that is acquired from the operating air conditioner **1**, and calculates an outlier. Specifically, the abnormality estimation model **46A** adopts, as normal sample values (the cluster that is classified as normal), the values of the second feature values that are used to generate the abnormality estimation model **46A**, and calculates an outlier that indicates a degree

of deviation from the normal sample values, for the detected values of the second feature values of each of the pairs that are acquired by the acquisition unit **41** of the operating air conditioner **1**. The outlier is obtained by quantifying a distance that indicates a degree of deviation from a boundary of the cluster that is classified as normal, and the degree of deviation increases with an increase in an absolute value of the quantified value. With an increase in the degree of deviation, the possibility that the detected value of the second feature value is abnormal increases.

(94) FIG. **11** is an explanatory diagram illustrating an example of abnormality detection by the outlier. The abnormality estimation unit **46** classifies the detected value of the second feature value as normal if the absolute value of the outlier of the detected value of the second feature value is, for example, smaller than an absolute value of “-150”, and classifies the detected value of the second feature value as abnormal if the absolute value of the outlier of the detected value of the second feature value is, for example, equal to or larger than the absolute value of “-150”. Meanwhile, an outlier threshold X is set to a certain value by which normal data is not erroneously detected as abnormal, based on a result of verification of values that are actually determined as abnormal in a collected failure history of the air conditioner **1**. If the absolute value of the calculated outlier is equal to or larger than the absolute value of the outlier threshold X, the abnormality estimation unit **46** classifies the detected value of the second feature value as abnormal.

(95) If the detected value of the second feature value is classified as abnormal, the abnormality estimation unit **46** does not cause the refrigerant amount estimation unit **45** to perform operation of estimating the refrigerant shortage rate by using the detected value of the first feature value that is detected at the same time as the detected value of the second feature value. Further, the abnormality estimation unit **46** stores the detected value of the second feature value that is classified as abnormal in the abnormality log storage unit **43A** as an abnormality log.

(96) If the absolute value of the estimated outlier is smaller than the absolute value of the outlier threshold X, the abnormality estimation unit **46** classifies the detected value of the second feature value as normal. In this case, the abnormality estimation unit **46** causes the refrigerant amount estimation unit **45** to perform operation of estimating the refrigerant shortage rate by using the detected value of the first feature value that is detected at the same time as the detected value of the second feature value. Meanwhile, the abnormality estimation unit **46** classifies the detected value of the second feature value as normal when only the refrigerant leaked state is changed.

(97) Meanwhile, for convenience of explanation, the case has been described in which the outlier threshold X is set to, for example, “-150”, but the threshold may be adjusted appropriately based on the result of verification of the values that are actually determined as abnormal in the collected failure history.

(98) FIG. **12** is an explanatory diagram illustrating an example of a determination result of the determination unit **46D**. The abnormality estimation unit **46** outputs an estimation result in which the detected value of the second feature value of each of the pairs P**1** to P**4** is classified as abnormal or normal. The determination unit **46D** stores the detected value of the second feature value of each of the pairs P**1** to P**4**. The determination unit **46D** determines whether the estimation results of the detected values of the second feature value of the pairs P**1** to P**4** include abnormality. When the detected value of the second feature value of each of the pairs P**1** to P**4** is abnormal, and, for example, if the detected values of the second feature values of all of the pairs P**1** to P**4** are abnormal, the determination unit **46D** determines that the abnormality of the refrigerant circuit **6** is caused by the outdoor unit **2** that is shared by all of the pairs P**1** to P**4**. When the detected value of the second feature value of each of the pairs P**1** to P**4** is abnormal, and if the detected value of the second feature value of only a certain pair is abnormal, the determination unit **46D** determines that the abnormality of the refrigerant circuit **6** is caused by the indoor unit **3** in the certain pair in which the abnormality has occurred.

(99) In FIG. **12**, for example, if the detected values of the second feature values of the pairs P**1**, P**2** and P**4** are normal and the detected value of the second feature value of the pair P**3** is abnormal, the

determination unit **46D** determines that the abnormality of the refrigerant circuit **6** is caused by the indoor unit **3C** of the pair **P3**. Further, although not illustrated in FIG. **12**, for example, if the detected values of the second feature values of the pair **P1** and the pair **P2** are normal and the detected values of the second feature values of the pair **3** and the pair **4** are abnormal, the determination unit **46D** determines that the abnormality of the refrigerant circuit **6** is caused by the indoor unit **3C** of the pair **P3** and the indoor unit **3D** of the pair **P4**.

(100) Operation of Estimation Process

(101) FIG. **13** is a flowchart illustrating an example of processing operation performed by the control circuit **19** in relation to the estimation process. Meanwhile, it is assumed that the refrigerant amount estimation unit **45** in the control circuit **19** stores therein the first cooler estimation model **45A1**, the second cooler estimation model **45A2**, the third cooler estimation model **45A3**, the first heater estimation model **45A4**, the second heater estimation model **45A5**, and the third heater estimation model **45A6** that are generated in advance. Further, it is assumed that the abnormality estimation unit **46** in the control circuit **19** stores therein the cooling-period abnormality estimation model **46B** and the heating-period abnormality estimation model **46C** that are generated in advance. The estimation process is periodically performed once in a predetermined time period (for example, night time) in one day with respect to operating state quantities that are sequentially detected every 10 minutes in 24 hours by the detection unit. Meanwhile, the night time is described as an example of the predetermined time period; however, for example, the operating state quantities corresponding to one day are acquired after operation of the air conditioner **1** is stopped in the night time that is a time period in which operating frequency of the air conditioner **1** is low. Further, as the predetermined time period, it is possible to determine a predetermined time in which operation is not performed, rather than the night time, by examining operating states of the air conditioner **1** for one month, for example.

(102) In FIG. **13**, the control unit **44** in the control circuit **19** collects the operating state quantities as pieces of operating data via the acquisition unit **41** (Step **S11**). The control unit **44** performs a data filtering process for extracting an arbitrary operating state quantity from among the collected pieces of operating data (Step **S12**). The control unit **44** performs a data cleansing process (Step **S13**). Further, the abnormality estimation unit **46** performs an abnormality estimation process for each of the pairs **P1** to **P4** for classifying whether the detected value of the second feature value that is subjected to the data cleansing process using the abnormality estimation model **46A** is normal or abnormal (Step **S14**). In the abnormality estimation process, a classification result indicating abnormal or normal is estimated for each of the pairs **P1** to **P4** by using the abnormality estimation model **46A**.

(103) The control unit **44** determines whether the detected value of the second feature value of each of the pairs **P1** to **P4** is abnormal (Step **S15**). If the detected value of the second feature value of each of the pairs **P1** to **P4** is not abnormal (Step **S15**: No), the abnormality estimation unit **46** performs a remaining refrigerant amount estimation process of applying the detected value of the first feature value, which is acquired at the same time as the detected value of the second feature value of a pair that is classified as normal, to each of the refrigerant amount estimation models (Step **S16**). Further, the refrigerant amount estimation unit **45** calculates the refrigerant shortage rate of the refrigerant circuit **6** (Step **S17**), and terminates the processing operation illustrated in FIG. **13**.

(104) Furthermore, if the detected value of the second feature value of each of the pairs **P1** to **P4** is abnormal (Step **S15**: Yes), the determination unit **46D** in the abnormality estimation unit **46** determines that the refrigerant circuit **6** is abnormal, and determines whether the detected values of the second feature values of all of the pairs **P1** to **P4** are abnormal (Step **S18**). If the detected values of the second feature values of all of the pairs **P1** to **P4** are abnormal (Step **S18**: Yes), the determination unit **46D** determines that the abnormality of the refrigerant circuit **6** is caused by abnormality of the outdoor unit **2** (Step **S19**). Then, the abnormality estimation unit **46** performs an

abnormality output process (Step S20), and terminates the processing operation illustrated in FIG.

13. As a result, the abnormality estimation unit 46 is able to identify that the abnormality of the refrigerant circuit 6 is caused by abnormality of the outdoor unit 2.

(105) If not all of the detected values of the second feature values of all of the pairs P1 to P4 are abnormal (Step S18: No), the determination unit 46D determines that the detected value of the second feature value of only a certain pair is abnormal (Step S21). Meanwhile, when the determination unit 46D determines that the detected value of the second feature value of only a certain pair is abnormal, it is possible to also identify a pair in which the abnormality has occurred as described above. Furthermore, if it is determined that the detected value of the second feature value of only a certain pair is abnormal, the determination unit 46D determines that the abnormality of the refrigerant circuit 6 is caused by abnormality of the indoor unit 3 of the certain pair that is determined as abnormal (Step S22), and returns to Step S20 to perform the abnormality output process. As a result, the abnormality estimation unit 46 is able to identify the indoor unit 3 that causes the abnormality of the refrigerant circuit 6 among the plurality of indoor units 3.

(106) In the data filtering process, not all of the operating state quantities are used, but only a part of the operating state quantities (the detected value of the first feature value and the detected value of the second feature value) that is needed for the abnormality estimation process or that is needed to calculate the refrigerant shortage rate from among the plurality of operating state quantities is extracted based on a predetermined filter condition. By assigning the first feature value and the detected value of the second feature value (an abnormal value and an outlier are eliminated) that are subjected to the data filtering process (to be described later) to the generated refrigerant amount estimation model 45A and the generated abnormality estimation model 46A, it is possible to more accurately estimate abnormality by using the second feature value and more accurately estimate the refrigerant shortage rate by using the first feature value.

(107) The predetermined filter condition includes a first filter condition, a second filter condition, and a third filter condition. The first filter condition is a filter condition for data that is extracted commonly among all of operating modes of the air conditioner 1, for example. The second filter condition is a filter condition for data that is extracted at the time of cooling operation. The third filter condition is a filter condition for data that is extracted at the time of heating operation.

(108) The first filter condition is, for example, a driving state of the compressor 11, identification of an operating mode, elimination of special operation, elimination of a missing value with respect to an acquired value, selection of a small value of a change amount of an operating state quantity that largely affects generation of each of the regression equations, or the like. The driving state of the compressor 11 is a condition needs to be determined because it is impossible to estimate the refrigerant shortage rate unless the compressor stably operates and the refrigerant circulates in the refrigerant circuit 6, and is the filter condition that is provided to eliminate an operating state quantity that is detected during a transition period, such as at the time of activation of the compressor 11.

(109) The identification of the operating mode is a filter condition for extracting only an operating state quantity that is acquired at the time of cooling operation and at the time of heating operation. Therefore, an operating state quantity that is acquired during dehumidification operation or air supply operation is eliminated. The elimination of the special operation is a filter condition for eliminating an operating state quantity that is acquired during special operation, such as oil recovery operation or defrosting operation, in which the state of the refrigerant circuit 6 largely differs from the state at the time of cooling operation and at the time of heating operation. The elimination of the missing value is a filter condition for eliminating an operating state quantity that includes a missing value because when the operating state quantity that is used for determination of the refrigerant shortage rate includes a missing value, and if the operating state quantity is used to generate each of the regression equations, accuracy of each of the regression equations may be reduced.

(110) The selection of the small value of the change amount of the operating state quantity that is assigned to each of the regression equations or each of the refrigerant shortage rate calculation formulas is a filter condition for extracting only an operating state quantity in a case where the operating state of the air conditioner **1** is stable, and is a condition that is needed to improve estimation accuracy using each of the regression equations and each of the refrigerant shortage rate calculation formulas. Meanwhile, the operating state quantity that has large influence is, for example, the degree of supercooling of the refrigerant that is used when the refrigerant shortage rate is 0% to 30% at the time of cooling operation, the suction temperature that is used when the refrigerant shortage rate is 40% to 70% at the time of cooling operation, the degree of suction superheat at the time of heating operation, or the like.

(111) The second filter condition includes, for example, elimination of the heat exchange outlet temperature, abnormality of the subcool, abnormality of the discharge temperature, or the like.

(112) The elimination of the heat exchange outlet temperature is a filter condition that takes into account the fact that, because the outdoor air temperature sensor **36** and the refrigerant temperature sensor **35** are located close to each other, the heat exchange outlet temperature detected by the refrigerant temperature sensor **35** at the time of cooling operation does not become lower than the outdoor air temperature detected by the outdoor air temperature sensor **36**, and is a filter condition for eliminating the heat exchange outlet temperature that is lower than the outdoor air temperature.

(113) The abnormality of the subcool is a filter condition for eliminating a degree of supercooling of the refrigerant that is abnormally high or abnormally low because a cooling load is extremely large or small when the degree of supercooling of the refrigerant as described above is detected. The abnormality of the discharge temperature is a filter condition for eliminating discharge temperature that is detected when what is called an out-of-gas state is detected in which the amount of refrigerant that is sucked into the compressor **11** is reduced due to a small cooling load.

(114) The third filter condition is, for example, abnormality of the discharge temperature or the like. When the discharge temperature increases due to a large heating load at the time of heating operation and discharge temperature protection control is performed, the rotation speed of the compressor **11** is reduced to reduce the discharge temperature, and, the third filter condition is a filter condition for eliminating the discharge temperature that is detected at this time.

(115) The data cleansing process is a process for eliminating the detected value of the first feature value that may lead to erroneous estimation, instead of using all of the acquired detected values of the first feature values for estimation of the refrigerant shortage rate. Further, the data cleansing process is also a process for eliminating the detected value of the second feature value that may lead to erroneous abnormality estimation, instead of using all of the acquired detected values of the second feature values for the abnormality estimation process. Specifically, the acquired operating state quantities may be smoothed to perform noise control, data amount limitation, or the like. The noise control based on the data smoothing is a process of preventing noise by calculating averages in a subject interval and calculating a moving average of the degree of supercooling of the refrigerant, the suction temperature, and the degree of suction superheat in each of the models, for example. The data amount limitation is a process for eliminating data whose amount is small because reliability of such data is low, for example. For example, if the number of pieces of data that remain after the filtering process is performed on pieces of input data corresponding to one day is equal to or larger than X, the data is used for estimation of the refrigerant shortage rate or the abnormality estimation process on the second feature value, and if the number of pieces of data is smaller than X, all pieces of the data corresponding to the day are not used. In other words, in the data cleansing process, it is possible to more accurately estimate the refrigerant shortage rate by assigning the operating state quantities, from which the abnormal value and the outlier are eliminated, to the refrigerant amount estimation model **45A**, and it is possible to more accurately estimate abnormality by assigning the operating state quantities, from which the abnormal value and the outlier are eliminated, to the abnormality estimation model **46A**.

(116) The abnormality estimation process is a process of calculating the degree of deviation (outlier) from a local maximum value (center of a cluster) of the density function based on the density function of an entire distribution that is estimated from simulation values of the second feature values, and determining whether the outlier falls within a predetermined range (whether determination target data is included in the cluster). The second feature value of each of the pairs **P1** to **P4** acquired from the operating air conditioner **1** is applied to the abnormality estimation model **46A** and an outlier is calculated. In the abnormality estimation process, the value of the second feature value that is used to generate the abnormality estimation model **46A** is adopted as a normal sample value, and an outlier from the normal sample value of the detected value of the second feature value of each of the pairs **P1** to **P4** that is acquired by the acquisition unit **41** at a different timing is calculated. Further, in the abnormality estimation process, if the absolute value of the calculated outlier is equal to or larger than the absolute value of the outlier threshold **X**, the detected value of the second feature value in the subject pair is classified as abnormal. Furthermore, in the abnormality estimation process, if the absolute value of the calculated outlier is smaller than the absolute value of the outlier threshold **X**, the detected value of the second feature value of the subject pair is classified as normal.

(117) The determination unit **46D** is able to identify the indoor unit **3** or the outdoor unit **2** as a cause of the abnormality of the refrigerant circuit **6** based on the classification result of each of the pairs **P1** to **P4**. If the detected values of the second feature values of all of the pairs **P1** to **P4** are abnormal, the determination unit **46D** identifies abnormality of the outdoor unit **2** as the cause of the abnormality of the refrigerant circuit **6**. Further, if the detected value of the second feature value of a certain pair is abnormal, the determination unit **46D** identifies abnormality of the indoor unit **3** of the certain pair that is classified as abnormal as the cause of the abnormality of the refrigerant circuit **6**.

(118) FIG. **14** is a flowchart illustrating an example of the processing operation performed by the control circuit **19** in relation to the remaining refrigerant amount estimation process. The estimation of the remaining refrigerant amount is a process of calculating the refrigerant shortage rate of the refrigerant circuit **6** at a current time by, for example, assigning the detected value of the first feature value, which is acquired at the same time as the detected value of the second feature value that is classified as normal in the abnormality estimation process among the current operating state quantities (sensor values) that are subjected to the data filtering process and the data cleansing process, to each of the regression equations or each of the refrigerant shortage rate calculation formulas of the refrigerant amount estimation model **45A**. In FIG. **14**, the refrigerant amount estimation unit **45** in the control circuit **19** determines whether the acquired first feature value is acquired during cooling operation (Step **S31**). If the acquired first feature value is acquired during cooling operation (Step **S31**: Yes), the refrigerant amount estimation unit **45** applies the first feature value to each of the first cooler estimation model **45A1** to the third cooler estimation model **45A3** (Step **S32**).

(119) If the acquired first feature value is not acquired during cooling operation (Step **S31**: No), that is, if the acquired first feature value is acquired during heating operation, the refrigerant amount estimation unit **45** applies the first feature value to each of the first heater estimation model **45A4** to the third heater estimation model **45A6** (Step **S33**). Further, the refrigerant amount estimation unit **45** calculates the refrigerant shortage rate at the current time by combining results obtained by applying the first feature value to each of the first cooler estimation model **45A1** to the third cooler estimation model **45A3** and results obtained by applying the first feature value to each of the first heater estimation model **45A4** to the third heater estimation model **45A6** (Step **S34**), and terminates the processing operation illustrated in FIG. **14**.

(120) In the abnormality output process, the detected value of the second feature value that is classified as abnormal in the abnormality estimation process is stored, as an abnormality log, in the abnormality log storage unit **43A** and an alarm is output. As a result, it is possible to store the

abnormal detected value of the second feature value.

Effect of First Embodiment

(121) In the air conditioner **1** of the first embodiment, the value of the second feature value that is used to generate the abnormality estimation model **46A** is adopted as a normal sample value, and an outlier from the normal sample value of the detected value of the second feature value of each of the pairs **P1** to **P4** that is detected at a different timing is calculated. Further, in the air conditioner **1**, if the absolute value of the calculated outlier is equal to or larger than the absolute value of the outlier threshold **X**, the detected value of the second feature value of the subject pair is classified as abnormal, and it is estimated that the refrigerant circuit **6** is abnormal. Furthermore, in the air conditioner **1**, the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value of the pair that is classified as abnormal is not used for the refrigerant amount estimation model **45A**. As a result, it is possible to accurately estimate the refrigerant shortage rate of the refrigerant circuit **6**.

(122) The air conditioner **1** is able to identify the indoor unit **3** or the outdoor unit **2** as a cause of the abnormality of the refrigerant circuit **6** based on the classification result of each of the pairs **P1** to **P4**. If the detected values of the second feature values of all of the pairs **P1** to **P4** are abnormal, the air conditioner **1** identifies abnormality of the outdoor unit **2** as the cause of the abnormality of the refrigerant circuit **6**. Further, if the detected value of the second feature value of a certain pair is abnormal, the air conditioner **1** identifies abnormality of the indoor unit **3** of the certain pair that is classified as abnormal as the cause of the abnormality of the refrigerant circuit **6**. As a result, even if it is estimated that abnormality other than a change of the remaining refrigerant amount has occurred, it is possible to estimate the outdoor unit **2** or the indoor unit **3** in which the abnormality has occurred.

(123) For example, when the refrigerant shortage rate is to be estimated by the refrigerant amount estimation model **45A** that is generated by a linear analysis of the multiple regression analysis, and if the first feature value has changed due to leakage of the refrigerant and a failure other than the leakage of the refrigerant, the refrigerant shortage rate may be estimated as a small value even if the refrigerant shortage rate is increased (=abnormal) depending on a degree of change of each of the feature values. For example, if the rotation speed of the compressor and the suction temperature are changed due to a failure other than the leakage of the refrigerant, and the amounts of changes of the respective values are cancelled out, the refrigerant shortage rate may be estimated as a small value (=a normal amount). However, in the air conditioner **1** of the present embodiment, the detected value of the first feature value, which is acquired at the same time as the detected value of the second feature value that is classified as abnormal by the abnormality estimation model **46A** that is generated by a non-linear analysis, such as the Kernel density estimation method, is not used. As a result, it is possible to prevent erroneous estimation of the refrigerant shortage rate.

(124) Furthermore, originally, if the refrigerant amount estimation model **45A** that is generated by the linear analysis is used, it may be possible to estimate that the refrigerant shortage rate has increased (=abnormal) even though the refrigerant shortage rate has a small value (=normal). For example, there may be a case in which the refrigerant shortage rate may be estimated as having been increased as a result of a change of the rotation speed of the compressor due to a failure other than the leakage of the refrigerant. However, in the air conditioner **1** of the first embodiment, the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is classified as abnormal by the abnormality estimation model **46A** that is generated by the non-linear analysis is not used for the refrigerant amount estimation model **45A**. As a result, it is possible to prevent erroneous estimation of the refrigerant shortage rate.

(125) If the absolute value of the calculated outlier is smaller than the absolute value of the outlier threshold **X**, the abnormality estimation model **46A** of the air conditioner **1** classifies the detected value of the second feature value of the subject pair as normal. Further, the air conditioner **1** performs the multiple regression analysis on the detected value of the first feature value that is

acquired at the same time as the detected value of the second feature value of the pair that is classified as normal, and calculates the refrigerant shortage rate of the refrigerant circuit **6**. As a result, it is possible to accurately estimate the refrigerant shortage rate of the refrigerant circuit **6**. (126) The abnormality estimation model **46A** that is mounted on the air conditioner **1** is generated by a non-linear analysis, such as the Kernel density estimation method, by using a part of the detected value of the first feature value used for the refrigerant amount estimation model **45A** and by using the value of the second feature value that includes an operating state quantity that largely affects cooling cycle operation. The abnormality estimation model **46A** classifies the detected value of the second feature value of each of the pairs **P1** to **P4** as normal or abnormal. Further, in the refrigerant amount estimation model **45A**, the refrigerant amount estimation model **45A** is generated by using the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is classified as normal, instead of using all of the operating state quantities. As a result, it is possible to generate the refrigerant amount estimation model **45A** with high accuracy.

(127) In the present embodiment, each of the regression equations of the refrigerant amount estimation model **45A** is generated by using the detected value of the first feature value that is obtained by a simulation, and the detected value of the first feature value that is obtained by the simulation does not include an abnormal value and a certain value that is extremely large or small as compared to other values. In this manner, the detected value of the operating state quantity that is subjected to the data filtering process and the data cleansing process to eliminate an abnormal value and an outlier is assigned to each of the regression equations or each of the refrigerant shortage rate calculation formulas of the refrigerant amount estimation model **45A** that is generated using the feature value that is obtained by a simulation. In this case, by assigning only the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is classified as normal by using the abnormality estimation model **46A**, it is possible to more accurately estimate the refrigerant shortage rate.

(128) The abnormality estimation model **46A** is generated by using the feature value that is obtained by a simulation, and the feature value that is obtained by the simulation does not include an abnormal value and a certain value that is extremely larger or smaller than other values. By applying the detected value of the second feature value from which the abnormal value and the outlier are eliminated by performing the data filtering process and the data cleansing process as described above to the abnormality estimation model **46A** that is generated by using the feature value that does not include the abnormal value and the outlier, it is possible to more accurately determine the detected value of the second feature value. Further, by performing the data filtering process and the data cleansing process, the control circuit **19** is able to reduce the amount of data used to calculate the outlier by the abnormality estimation model **46A**, so that it is possible to reduce a time needed for the calculation of the outlier by the abnormality estimation model **46A** and reduce a load on the control circuit **19**.

(129) Meanwhile, in the first embodiment as described above, the example has been described in which the simulation result of each of the operating state quantities is obtained at the design stage of the air conditioner **1**, and the control circuit **19** stores the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** that are obtained by causing an information processing apparatus, such as a server, with a learning function to learn a simulation result. Alternatively, it may be possible to provide a server **120** that is connected to the air conditioner **1** by a communication network **110**, and cause the server **120** to generate the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** and transmit an estimation result of the refrigerant amount estimation model **45A** and an estimation result of the abnormality estimation model **46A** to the air conditioner **1**. This embodiment will be described below.

Second Embodiment

(130) Configuration of Air Conditioning System

(131) FIG. 15 is an explanatory diagram illustrating an example of an air conditioning system 100 of a second embodiment. Meanwhile, the same components as those of the air conditioner 1 of the first embodiment are denoted by the same reference symbols, and explanation of the same components and operation will be omitted. The air conditioning system 100 illustrated in FIG. 15 includes the air conditioner 1, the communication network 110, and the server 120. The air conditioner 1 includes the compressor 11, the outdoor unit 2 that includes the outdoor heat exchanger 13 and the outdoor unit expansion valve 14, the indoor unit 3 that includes the indoor heat exchanger 51, and a control circuit 19A. The air conditioner 1 includes the refrigerant circuit 6 that is configured by connecting the outdoor unit 2 and the indoor unit 3 by refrigerant pipes, such as the liquid pipe 4 and the gas pipe 5, and a predetermined amount of refrigerant is stored in the refrigerant circuit 6. The control circuit 19A includes the acquisition unit 41, the communication unit 42, the storage unit 43, and the control unit 44. Meanwhile, the control circuit 19A does not include the refrigerant amount estimation unit 45, the abnormality estimation unit 46, and the abnormality log storage unit 43A.

(132) The server 120 includes a generation unit 121, a communication unit 121A, a refrigerant amount estimation unit 122, an abnormality estimation unit 123, and a storage unit 124. The storage unit 124 includes an abnormality log storage unit 124A. The generation unit 121 generates the refrigerant amount estimation model 45A by a multiple regression analysis method by using a detected value or a simulation value of the first feature value related to estimation of the refrigerant shortage rate of the refrigerant that is stored in the refrigerant circuit 6. Meanwhile, the refrigerant amount estimation model 45A includes, for example, the first cooler estimation model 45A1, the second cooler estimation model 45A2, the third cooler estimation model 45A3, the first heater estimation model 45A4, the second heater estimation model 45A5, and the third heater estimation model 45A6 that are explained in the first embodiment. The refrigerant amount estimation unit 122 stores therein the refrigerant amount estimation model 45A that is generated by the generation unit 121. Further, the generation unit 121 generates the abnormality estimation model 46A by the Kernel density estimation method by using the detected values of the second feature values of all of the pairs P1 to P4 that are obtained by a simulation in the steady state and the refrigerant leaked state. Meanwhile, the abnormality estimation model 46A includes, for example, the cooling-period abnormality estimation model 46B and the heating-period abnormality estimation model 46C described in the first embodiment.

(133) The abnormality estimation unit 123 stores therein the abnormality estimation model 46A that is generated by the generation unit 121. The abnormality estimation unit 123 classifies the detected value of the second feature value as normal or abnormal by using the abnormality estimation model 46A. If the detected value of the second feature value is classified as abnormal, the abnormality estimation unit 123 stores the detected value of the second feature value that is classified as abnormal, as an abnormality log, in the abnormality log storage unit 124A. Further, the determination unit 46D in the abnormality estimation unit 123 identifies the indoor unit 3 or the outdoor unit 2 that is a cause of the abnormality of the refrigerant circuit 6 based on a classification result obtained by the abnormality estimation unit 123, that is, a classification result of each of the pairs P1 to P4. The communication unit 121A transmits a result of identification of the indoor unit 3 or the outdoor unit 2 as the cause of the abnormality of the refrigerant circuit 6, which is obtained by the determination unit 46D, to the air conditioner 1 via the communication network 110. The control circuit 19A of the air conditioner 1 is able to identify the cause of the abnormality of the refrigerant circuit 6 based on the result of identification of the indoor unit 3 or the outdoor unit 2 as the cause of the abnormality of the refrigerant circuit 6, which is received from the server 120.

(134) Furthermore, the refrigerant amount estimation unit 122 calculates the refrigerant shortage rate in the refrigerant circuit 6 of the air conditioner 1 by using the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is classified as normal by the abnormality estimation model 46A and by using the received

refrigerant amount estimation model **45A**. The communication unit **121A** transmits the refrigerant shortage rate that is calculated by the refrigerant amount estimation unit **122** to the air conditioner **1** via the communication network **110**. The control circuit **19A** of the air conditioner **1** is able to identify the refrigerant shortage rate of the refrigerant circuit **6** based on the refrigerant shortage rate that is received from the server **120**.

(135) The generation unit **121** generates or updates the cooling-period abnormality estimation model **46B** by using the values of the second feature values of all of the pairs **P1** to **P4** that are obtained by a simulation in the steady state and the refrigerant leaked state at the time of cooling while the refrigerant circuit **6** is in the normal state.

(136) The generation unit **121** periodically collects the operating state quantities at the time of cooling operation from a standard machine (installed in a test room or the like of a manufacturing company) of the air conditioner **1** that is able to measure the steady state and the refrigerant leaked state at the time of cooling while the refrigerant circuit **6** is in the normal state, and generates or updates the cooling-period abnormality estimation model **46B** by using a comparison result between a classification result indicating normal or abnormal obtained by the cooling-period abnormality estimation model **46B** and an actually measured classification result and by using the collected operating state quantities. As a result, it is possible to generate the cooling-period abnormality estimation model **46B** with high accuracy.

(137) The generation unit **121** periodically collects the operating state quantities at the time of cooling operation from the standard machine (installed in the test room or the like of the manufacturing company) of the air conditioner **1** that is able to measure the refrigerant shortage rate of the refrigerant circuit **6**, and generates or updates the first cooler estimation model **45A1**, the second cooler estimation model **45A2**, and the third cooler estimation model **45A3** by using a comparison result between the refrigerant shortage rate that is estimated by each of the refrigerant amount estimation models **45A** and the actually measured refrigerant shortage rate and by using the collected operating state quantities. Meanwhile, as in the first embodiment, it may be possible to obtain, by a simulation, the operating state quantity that is used to generate each of the refrigerant amount estimation models **45A**, and the generation unit **121** may generate each of the refrigerant amount estimation models **45A** by using each of the operating state quantities that are obtained by the simulation.

(138) The generation unit **121** generates or updates the heating-period abnormality estimation model **46C** by using the values of the second feature values of all of the pairs **P1** to **P4** that are obtained by a simulation in the steady state and the refrigerant leaked state at the time of heating while the refrigerant circuit **6** is in the normal state.

(139) The generation unit **121** periodically collects the operating state quantities at the time of heating operation from the standard machine (installed in the test room or the like of the manufacturing company) of the air conditioner **1** that is able to measure the steady state and the refrigerant leaked state at the time of heating while the refrigerant circuit **6** is in the normal state, and generates or updates the heating-period abnormality estimation model **46C** by using a comparison result between the classification result indicating normal or abnormal obtained by the heating-period abnormality estimation model **46C** and the actually measured classification result and by using the collected operating state quantities. As a result, it is possible to generate the heating-period abnormality estimation model **46C** with high accuracy.

(140) The generation unit **121** periodically collects the operating state quantities at the time of heating operation from the standard machine of the air conditioner **1** as described above, and generates the first heater estimation model **45A4**, the second heater estimation model **45A5**, and the third heater estimation model **45A6** by using a comparison result between the refrigerant shortage rate that is estimated by each of the refrigerant amount estimation models **45A** and by using the collected operating state quantities. Meanwhile, as in the first embodiment, it may be possible to obtain, by a simulation, the operating state quantity that is used to generate each of the

refrigerant amount estimation models **45A**, and the generation unit **121** may generate each of the refrigerant amount estimation models **45A** by using the operating state quantity that is obtained by the simulation.

(141) The generation unit **121** generates the abnormality estimation model **46A** by using the feature value that is obtained by the simulation, and the value of the feature value that is obtained by the simulation does not include an abnormal value and an extremely large or small value as compared to other values. By applying the detected value of the second feature value from which the abnormal value and the outlier are eliminated by performing the data filtering process and the data cleansing process as described above to the abnormality estimation model **46A** that is generated by using the feature value that does not include the abnormal value and the outlier, it is possible to more accurately determine the detected value of the second feature value. Further, if the generation unit **121** performing the data filtering process and the data cleansing process on the second feature value as described in the first embodiment, it is possible to reduce the amount of data used to calculate the outlier by the abnormality estimation model **46A**. As a result, it is possible to reduce a time needed for the calculation of the outlier by the abnormality estimation model **46A** and reduce a usage rate of the server **120**; therefore, if the server **120** adopts a metered system in which costs increases with an increase in use, it is possible to reduce cost needed for the calculation of the outlier.

Effects of Second Embodiment

(142) The server **120** of the second embodiment generates the abnormality estimation model **46A** by using the values of the second feature values of all of the pairs **P1** to **P4** that are obtained by a simulation in the steady state and the refrigerant leaked state while the refrigerant circuit **6** is in the normal state, and stores the generated abnormality estimation model **46A** in the abnormality estimation unit **123**. The abnormality estimation unit **123** in the server **120** is able to classify whether the detected value of the second feature value of each of the pairs **P1** to **P4** obtained at a different timing is normal or abnormal by using the stored abnormality estimation model **46A**. Further, the air conditioner **1** estimates whether the refrigerant circuit **6** of each of the pairs **P1** to **P6** is abnormal or normal based on the classification result of the detected value of the second feature value of each of the pairs. The abnormality estimation unit **123** identifies the outdoor unit **2** or the indoor unit **3** as the cause of the abnormality of the refrigerant circuit **6** based on the estimation result indicating occurrence of abnormality of the refrigerant circuit **6** of each of the pairs. The communication unit **121A** transmits an identification result indicating the outdoor unit **2** or the indoor unit **3** as the cause of the abnormality of the refrigerant circuit **6** to the air conditioner **1**. As a result, the air conditioner **1** is able to identify the outdoor unit **2** or the indoor unit **3** as the cause of the abnormality of the refrigerant circuit **6**.

(143) The server **120** generates the refrigerant amount estimation model **45A** by using the value of the first feature value that is acquired from the air conditioner **1**, and stores the generated refrigerant amount estimation model **45A** in the refrigerant amount estimation unit **122**. The server **120** estimates the refrigerant shortage rate by using the stored refrigerant amount estimation model **45A**, and transmits the estimation result to the air conditioner **1** via the communication network **110**. As a result, the air conditioner **1** is able to recognize the refrigerant shortage rate of the refrigerant circuit **6**.

(144) Meanwhile, in the air conditioner **1** of the first embodiment and the second embodiment, the examples are described in which the four indoor units **3** are connected to the single outdoor unit **2**, but the number of the indoor units **3** is not limited to four as long as the plurality of indoor units **3** are provided, and the number may be changed appropriately.

(145) Further, in the present embodiment, the case has been described in which a relative refrigerant amount is estimated as an amount that represents the amount of refrigerant that remains in the refrigerant circuit **6**. Specifically, the case has been described in which the refrigerant shortage rate that is a ratio of an amount of refrigerant that has leaked to the outside from the

refrigerant circuit **6** to the storage amount (initial amount) of the refrigerant that is stored in the refrigerant circuit **6** is estimated and provided. However, the present invention is not limited to this example, and it may be possible to multiply the estimated refrigerant shortage rate by the initial value, and provide the amount of the refrigerant that has leaked to the outside from the refrigerant circuit **6**. Furthermore, it may be possible to generate an estimation model for estimating an absolute amount of the refrigerant that has leaked to the outside from the refrigerant circuit **6** or an absolute amount of the refrigerant that remains in the refrigerant circuit **6**, and provide an estimation result that is obtained by the estimation model. When the estimation model for estimating the absolute amount of the refrigerant that has leaked to the outside from the refrigerant circuit **6** or the absolute amount of the refrigerant that remains in the refrigerant circuit **6** is to be generated, it is sufficient to take into account volumes of the outdoor heat exchanger **13** and each of the indoor heat exchangers **51** and a volume of the liquid pipe **4** as described above, in addition to each of the operating state quantities as described above.

Modification

(146) Meanwhile, in the present embodiment, for example, the case has been described in which the estimation result obtained by the first cooler estimation model **45A1** and the estimation result obtained by the second cooler estimation model **45A2** are interpolated by the sigmoid coefficient, but embodiments are not limited to the sigmoid coefficient; for example, an interpolation method, such as linear interpolation, may be used, and an appropriate change may be made.

(147) In the present embodiment, a part of simulation results among a plurality of simulation results are used, rather than using all of the simulation results. For example, the first cooler estimation model **45A1** that is used when the refrigerant shortage rate is 0% to 30% at the time of cooling operation, the second cooler estimation model **45A2** that is used when the refrigerant shortage rate is 40% to 70%, and the third cooler estimation model **45A3** that is used when the refrigerant shortage rate is 30% to 40% are generated separately. Therefore, the operating state quantities are prepared by simulations, so that when the operating quantities are collected by operating the air conditioner **1**, it is possible to easily collect a needed amount of operating state quantities by comparison.

(148) In the present embodiment, the case has been described in which the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** are generated by the server **120** or the control circuit **19**, but a user may calculate the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** from the simulation result. Further, in the present embodiment, the case has been described in which each of the estimation models is generated by using the multiple regression analysis method, but it may be possible to generate an estimation model by using support vector regression (SVR), a neural network (NN), or the like of a machine learning model that can perform a general regression analysis method. In this case, to select a feature value, it is sufficient to use a general method (a forward feature selection method, a backward feature elimination, or the like) for selecting a feature value such that accuracy of the estimation model is improved, instead of the P value and the correction value R² that are used in the multiple regression analysis method.

(149) The case has been described in which the abnormality estimation model **46A** is generated by using the values of the second feature values of all of the pairs that are obtained by a simulation in the steady state and the refrigerant leaked state while the refrigerant circuit **6** is in the normal state, and the outlier is calculated by adopting, as normal sample values, the values of the second feature values of all of the pairs and quantifying a distance between the detected value of the second feature value and the normal sample value in each of the pairs. However, the abnormality estimation model **46A** may be generated by using the value of the second feature value of each of the pairs that is obtained by a simulation in the steady state and the refrigerant leaked state while the refrigerant circuit **6** is in the normal state, and calculate the outlier by adopting, as the normal sample value, the value of the second feature value of each of the pairs used for the generation and

quantifying a distance from the detected value of the second feature value and the normal sample value in the same pair, where an appropriate change may be made.

(150) Furthermore, the case has been described in which the abnormality estimation model **46A** is generated by using the value of the second feature value that is obtained by a simulation in the steady state and the refrigerant leaked state while the refrigerant circuit **6** is in a normal state, but it may be possible to generate the model by using only the value of the second feature value that is obtained by a simulation in the steady state while the refrigerant circuit **6** is in the normal state, without using the value of the second feature value in the refrigerant leaked state.

(151) Moreover, in the present embodiment, the case has been described in which the abnormality estimation model **46A** is generated by using the Kernel density estimation method, but embodiments are not limited to the Kernel density estimation method as long as the method is a non-linear analytic method, and an appropriate change may be made.

(152) Furthermore, in the present embodiment, the example has been described in which the one or more indoor units **3** are connected to the single outdoor unit **2** in the air conditioner **1**, but the technology is applicable to the air conditioner **1** in which the one or more indoor units **3** are connected to the two or more outdoor units **2**.

(153) In the first embodiment, the case has been described in which the simulation result of each of the operating state quantities is obtained at the design stage of the air conditioner **1**, and the control circuit **19** stores therein the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** that are generated by causing an information processing apparatus, such as a server, with a learning function to learn a simulation result. However, it may be possible to provide a server that is connected to the air conditioner **1** via the communication network, and the server may generate and transmit the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** to the air conditioner **1**. Further, the air conditioner **1** may store the refrigerant amount estimation model **45A** and the abnormality estimation model **46A** that are received from the server in the control circuit **19**.

(154) The refrigerant circuit **6** is configured such that at least the one indoor unit **3**, which is connected to at least the one outdoor unit **2**, is connected by a refrigerant pipe. Therefore, the refrigerant amount estimation model **45A** is able to estimate the refrigerant shortage rate by using the detected value of the first feature value of the single representative outdoor unit **2** among at least the one outdoor unit **2** and the detected value of the first feature value of the single representative indoor unit **3** among at least the one indoor unit **3**. Meanwhile, it is assumed that the representative outdoor unit **2** is selected from at least the one operating outdoor unit **2** based on an arbitrary rule, and the representative indoor unit **3** is selected from at least the one operating indoor unit **3** based on an arbitrary rule. The arbitrary rule is, for example, ascending order of identification numbers that are assigned to respective devices.

(155) Furthermore, the components illustrated in the drawings need not necessarily be physically configured in the manner illustrated in the drawings. In other words, specific forms of distribution and integration of the components are not limited to those illustrated in the drawings, and all or part of the components may be functionally or physically distributed or integrated in arbitrary units depending on various loads or use conditions.

(156) Moreover, all or part of various processing functions implemented by each of the apparatuses may be implemented by a central processing unit (CPU) (or a microcomputer, such as a micro processing unit (MPU) or a micro controller unit (MCU)). Furthermore, all or part of each of the various processing functions may be realized by a CPU and a program analyzed and executed by the CPU, or may be realized by hardware using wired logic.

(157) Moreover, in each of the embodiments as described above, the refrigerant shortage rate is assumed as an amount of reduction from a prescribed amount, where 100% indicates that the prescribed amount of the refrigerant is stored. Alternatively, it may be possible to estimate the refrigerant shortage rate by the method described in the embodiments immediately after the

prescribed amount of the refrigerant is stored in the refrigerant circuit **6**, and adopt the estimation result as 100%. For example, if the refrigerant shortage rate that is estimated immediately after the prescribed amount of refrigerant is stored in the refrigerant circuit **6** is 90%, that is, if it is estimated that the amount of refrigerant that is currently stored in the refrigerant circuit **6** is smaller than the prescribed amount by 10%, it may be possible to adopt a refrigerant amount that is smaller than the prescribed amount by 10% as 100%. By adjusting the refrigerant amount that is adopted as 100% in accordance with the refrigerant amount, it is possible to more accurately estimate the refrigerant shortage rate from next time.

REFERENCE SIGNS LIST

(158) **1** air conditioner **2** outdoor unit **3** indoor unit **41** acquisition unit **44** control unit **45** refrigerant amount estimation unit **45A** refrigerant amount estimation model **46** abnormality estimation unit **46A** abnormality estimation model **46B** cooling-period abnormality estimation model **46C** heating-period abnormality estimation model **46D** determination unit **100** air conditioning system **120** server **121** generation unit **121A** communication unit **122** refrigerant amount estimation unit **123** abnormality estimation unit

Claims

1. An air conditioning system comprising: an air conditioner that includes a refrigerant circuit in which at least one or more indoor units are connected to an outdoor unit by refrigerant pipes; and a server that is communicably connected to the air conditioner, wherein the air conditioner includes: an first memory; and first processing circuitry coupled to the first memory and configured to: detect a state quantity related to control on the air conditioner; acquire a detected value of the state quantity that is detected by the detecting; and transmit the detected value acquired by the acquiring to the server, and the server includes: a second memory; and second processing circuitry coupled to the second memory and configured to: receive the detected value from the air conditioner; and estimate occurrence of abnormality of the refrigerant circuit by using a detected value of a feature value by assuming that the state quantity related to abnormality of the refrigerant circuit is adopted as the feature value, wherein the estimating adopts the outdoor unit and each of the indoor units as a single pair, estimates occurrence of abnormality in the refrigerant circuit for each of the pairs, estimates that abnormality has occurred in the indoor unit of a subject pair when estimating that abnormality has occurred in any of the pairs, and estimates that abnormality has occurred in the outdoor unit when estimating that abnormality has occurred in all of the pairs.
2. The air conditioning system according to claim 1, wherein a predetermined amount of refrigerant is stored in the refrigerant circuit, and the estimating estimates that the refrigerant circuit is normal when only a remaining refrigerant amount is changed.
3. The air conditioning system according to claim 1, wherein the second processing circuitry is further configured to estimate a remaining refrigerant amount of the refrigerant circuit by using a detected value of a first feature value by assuming that the state quantity related to a refrigerant amount of the refrigerant circuit is adopted as the first feature value, and the estimating estimates occurrence of abnormality of the refrigerant circuit by using a detected value of a second feature value by assuming that a state quantity that includes at least one state quantity that is included in the first feature value and at least one state quantity that is not included in the first feature value is adopted as the second feature value.
4. The air conditioning system according to claim 3, wherein the first estimating includes a refrigerant amount estimation model that is generated by using the first feature value, and estimates the remaining refrigerant amount by applying the detected value of the first feature value to the refrigerant amount estimation model, and the estimating includes an abnormality estimation model that is generated by using the second feature value, and estimates occurrence of abnormality of the refrigerant circuit by applying the detected value of the second feature value to the abnormality

estimation model.

5. The air conditioning system according to claim 4, wherein the abnormality estimation model adopts the second feature value that is used to generate the abnormality estimation model as a normal sample value and calculates an outlier that indicates a degree of deviation from the normal sample value with respect to the detected value of the second feature value that is acquired by the acquiring, and the estimating the abnormality estimates that abnormality has occurred in the refrigerant circuit when an absolute value of the outlier calculated by the abnormality estimation model is equal to or larger than a predetermined threshold, and estimates that the refrigerant circuit is normal when the absolute value of the outlier calculated by the abnormality estimation model is smaller than the predetermined threshold.

6. The air conditioning system according to claim 5, wherein only when the estimating the abnormality estimates that the refrigerant circuit is normal, the estimating the refrigerant amount estimates the remaining refrigerant amount of the refrigerant circuit by using the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is used when the refrigerant circuit is estimated as normal.

7. The air conditioning system according to claim 6, wherein the estimating the abnormality estimates occurrence of abnormality of the refrigerant circuit before the estimating the refrigerant amount estimates the remaining refrigerant amount.

8. The air conditioning system according to claim 4, wherein the second feature value is a state quantity that is obtained as a result of a simulation of operation of the refrigerant circuit when the refrigerant circuit operates normally and only the remaining refrigerant amount is changed.

9. The air conditioning system according to claim 4, wherein the refrigerant amount estimation model is generated by using a linear analysis, and the abnormality estimation model is generated by using a non-linear analysis.

10. The air conditioning system according to claim 3, wherein the second feature value is a state quantity that is obtained as a result of a simulation of operation of the refrigerant circuit when the refrigerant circuit operates normally and only the remaining refrigerant amount is changed.

11. An abnormality estimation method that is implemented by an air conditioning system that includes an air conditioner that includes a refrigerant circuit in which at least one or more indoor units are connected to an outdoor unit by refrigerant pipes; and a server that is connected to the air conditioner by communication, the abnormality estimation method comprising: detecting, by first processing circuitry of the air conditioner, a state quantity related to control on the air conditioner; acquiring, by the first processing circuitry of the air conditioner, a detected value of the detected state quantity; transmitting, by the first processing circuitry of the air conditioner, the acquired detected value to the server; receiving, by second processing circuitry of the server, the detected value from the air conditioner; estimating, by the second processing circuitry, occurrence of abnormality of the refrigerant circuit for each of pairs by using a detected value of a feature value by assuming that the state quantity related to abnormality of the refrigerant circuit is adopted as the feature value and by adopting the outdoor unit and each of the indoor units as a single pair; estimating, by the second processing circuitry, that abnormality has occurred in the refrigerant circuit of a subject pair when estimating that abnormality has occurred in any of the pair; and estimating, by the second processing circuitry, that abnormality has occurred in the outdoor unit when estimating that abnormality has occurred in all of the pairs.

12. The abnormality estimation method according to claim 11, wherein a predetermined amount of refrigerant is stored in the refrigerant circuit, and the estimating the abnormality includes estimating that the refrigerant circuit is normal when only a remaining refrigerant amount is changed.

13. The abnormality estimation method according to claim 11, further including: estimating, by the second processing circuitry, a remaining refrigerant amount of the refrigerant circuit by using a detected value of a first feature value by assuming that the state quantity related to a refrigerant

amount of the refrigerant circuit is adopted as the first feature value, wherein the estimating the abnormality includes estimating occurrence of abnormality of the refrigerant circuit by using a detected value of a second feature value by assuming that a state quantity that includes at least one state quantity that is included in the first feature value and at least one state quantity that is not included in the first feature value is adopted as the second feature value.

14. The abnormality estimation method according to claim 13, further including: a refrigerant amount estimation model that is generated by using the first feature value; and an abnormality estimation model that is generated by using the second feature value, wherein the estimating the refrigerant amount includes estimates the remaining refrigerant amount by applying the detected value of the first feature value to the refrigerant amount estimation model, and the estimating the abnormality includes estimating occurrence of abnormality of the refrigerant circuit by applying the detected value of the second feature value to the abnormality estimation model.

15. The abnormality estimation method according to claim 14, wherein the abnormality estimation model adopts the second feature value that is used to generate the abnormality estimation model as a normal sample value and calculates an outlier that indicates a degree of deviation from the normal sample value with respect to the detected value of the second feature value that is acquired by the acquiring, and the estimating the abnormality includes estimating that abnormality has occurred in the refrigerant circuit when an absolute value of the outlier calculated by the abnormality estimation model is equal to or larger than a predetermined threshold, and estimating that the refrigerant circuit is normal when the absolute value of the outlier calculated by the abnormality estimation model is smaller than the predetermined threshold.

16. The abnormality estimation method according to claim 15, wherein the estimating the refrigerant amount includes estimating, only when the estimating the abnormality estimates that the refrigerant circuit is normal, the remaining refrigerant amount of the refrigerant circuit by using the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is used when the refrigerant circuit is estimated as normal.

17. The abnormality estimation method according to claim 16, wherein the estimating the abnormality is performed before the estimating the refrigerant amount.

18. An air conditioner that includes a refrigerant circuit in which at least one or more indoor units are connected to an outdoor unit by refrigerant pipes, the air conditioner comprising: a first memory; and first processing circuitry coupled to the first memory and configured to: detect a state quantity related to control on the air conditioner; acquire a detected value of the state quantity that is detected by the detecting; and estimate occurrence of abnormality of the refrigerant circuit by using a detected value of a feature value by assuming that the state quantity related to abnormality of the refrigerant circuit is adopted as the feature value, wherein the estimating the abnormality adopts the outdoor unit and each of the indoor units as a single pair, estimates occurrence of abnormality in the refrigerant circuit for each of the pairs, estimates that abnormality has occurred in the indoor unit of a subject pair when estimating that abnormality has occurred in any of the pairs, and estimates that abnormality has occurred in the outdoor unit when estimating that abnormality has occurred in all of the pairs.

19. The air conditioner according to claim 18, wherein a predetermined amount of refrigerant is stored in the refrigerant circuit, and the estimating the abnormality estimates that the refrigerant circuit is normal when only a remaining refrigerant amount is changed.

20. The air conditioner according to claim 18, the first processing circuitry is further configured to estimate a remaining refrigerant amount of the refrigerant circuit by using a detected value of a first feature value by assuming that the state quantity related to a refrigerant amount of the refrigerant circuit is adopted as the first feature value, wherein the estimating the abnormality estimates occurrence of abnormality of the refrigerant circuit by using a detected value of a second feature value by assuming that a state quantity that includes at least one state quantity that is included in the first feature value and at least one state quantity that is not included in the first feature value is

adopted as the second feature value.

21. The air conditioner according to claim 20, wherein the estimating the refrigerant amount includes a refrigerant amount estimation model that is generated by using the first feature value, and estimates the remaining refrigerant amount by applying the detected value of the first feature value to the refrigerant amount estimation model, the estimating the abnormality includes an abnormality estimation model that is generated by using the second feature value, and estimates occurrence of abnormality of the refrigerant circuit by applying the detected value of the second feature value to the abnormality estimation model.

22. The air conditioner according to claim 21, wherein the abnormality estimation model adopts the second feature value that is used to generate the abnormality estimation model as a normal sample value and calculates an outlier that indicates a degree of deviation from the normal sample value with respect to the detected value of the second feature value that is acquired by the acquiring, and the estimating the abnormality estimates that abnormality has occurred in the refrigerant circuit when an absolute value of the outlier calculated by the abnormality estimation model is equal to or larger than a predetermined threshold, and estimates that the refrigerant circuit is normal when the absolute value of the outlier calculated by the abnormality estimation model is smaller than the predetermined threshold.

23. The air conditioner according to claim 22, wherein only when the estimating the abnormality estimates that the refrigerant circuit is normal, the estimating the refrigerant amount estimates the remaining refrigerant amount of the refrigerant circuit by using the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is used when the refrigerant circuit is estimated as normal.

24. The air conditioner according to claim 23, wherein the estimating the abnormality estimates occurrence of abnormality of the refrigerant circuit before the estimating the refrigerant amount estimates the remaining refrigerant amount.

25. The air conditioner according to claim 21, wherein the refrigerant amount estimation model is generated by using a linear analysis, and the abnormality estimation model is generated by using a non-linear analysis.

26. The air conditioner according to claim 20, wherein the second feature value is a state quantity that is obtained as a result of a simulation of operation of the refrigerant circuit when the refrigerant circuit operates normally and only the remaining refrigerant amount is changed.

27. An abnormality estimation method that is implemented by an air conditioner that includes a refrigerant circuit in which at least one or more indoor units are connected to an outdoor unit by refrigerant pipes, the abnormality estimation method comprising: detecting a state quantity related to control on the air conditioner; acquiring a detected value of the detected state quantity; estimating occurrence of abnormality of the refrigerant circuit by using a detected value of a first feature value by assuming that the state quantity related to a refrigerant amount of the refrigerant circuit is adopted as the first feature value; adopting the outdoor unit and each of the indoor units as a single pair; estimating occurrence of abnormality in the refrigerant circuit for each of the pairs; estimating that abnormality has occurred in the refrigerant circuit of a subject pair when estimating that abnormality has occurred in any of the pairs; and estimating that abnormality has occurred in the outdoor unit when estimating that abnormality has occurred in all of the pairs.

28. The abnormality estimation method according to claim 27, wherein a predetermined amount of refrigerant is stored in the refrigerant circuit, and the estimating the abnormality includes estimating that the refrigerant circuit is normal when only a remaining refrigerant amount is changed.

29. The abnormality estimation method according to claim 27, further including: estimating a remaining refrigerant amount of the refrigerant circuit by using a detected value of a first feature value by assuming that the state quantity related to a refrigerant amount of the refrigerant circuit is adopted as the first feature value, wherein the estimating the abnormality includes estimating

occurrence of abnormality of the refrigerant circuit by using a detected value of a second feature value by assuming that a state quantity that includes at least one state quantity that is included in the first feature value and at least one state quantity that is not included in the first feature value is adopted as the second feature value.

30. The abnormality estimation method according to claim 29, further including: a refrigerant amount estimation model that is generated by using the first feature value; and an abnormality estimation model that is generated by using the second feature value, wherein the estimating the refrigerant amount includes estimating the remaining refrigerant amount by applying the detected value of the first feature value to the refrigerant amount estimation model, and the estimating the abnormality includes estimating occurrence of abnormality of the refrigerant circuit by applying the detected value of the second feature value to the abnormality estimation model.

31. The abnormality estimation method according to claim 30, wherein the abnormality estimation mode adopts the second feature value that is used to generate the abnormality estimation model as a normal sample value and calculates an outlier that indicates a degree of deviation from the normal sample value with respect to the acquired detected value of the second feature value, and the estimating the abnormality includes estimating occurrence of abnormality in the refrigerant circuit when an absolute value of the outlier calculated by the abnormality estimation model is equal to or larger than a predetermined threshold, and estimating that the refrigerant circuit is normal when the absolute value of the outlier calculated by the abnormality estimation model is smaller than the predetermined threshold.

32. The abnormality estimation method according to claim 31, wherein the estimating the refrigerant amount includes estimating, only when the estimating the abnormality estimates that the refrigerant circuit is normal, the remaining refrigerant amount of the refrigerant circuit by using the detected value of the first feature value that is acquired at the same time as the detected value of the second feature value that is used when the refrigerant circuit is estimated as normal.

33. The abnormality estimation method according to claim 32, wherein the estimating the abnormality is performed before the estimating the refrigerant amount.
